BEYOND THE NEXT FOLD: CP2SEQ FOR LONG-HORIZON PLANNING WITH EXACT VERIFICATION

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ABSTRACT

Many planning tasks require long action sequences whose consequences compound. We introduce **CP2Seq**, a benchmark of 600 generated instances in which a model receives a crease pattern and a target folded stack and constructs an executable sequence from an unfolded square. An environment checks each action; terminal scoring checks both creases and the stack, including layer order and original-sheet region identity under stated numerical tolerances. We evaluate Luna models with tool and reasoning-effort ablations and add a 20-instance Sonnet evaluation. An identity audit exposes a gap in the original plane-only scoring: five of 55 solves in the 360-case ablations fail the stronger identity check. Matched-instance search baselines show that low model success does not imply unreachable targets: random legal folding succeeds in 57% of easy and 96% of some-layers episodes, and BFS solves five of ten medium instances. These comparisons use explicit, differing resource and stopping budgets. Removing images does not reduce solve rate when exact numeric layer descriptions remain. An automatic stopping audit adds only two solves among 143 logged episodes, while repeated trials and backtracking analyses quantify variance and observable search behavior. We do not infer internal failure causes from termination labels. Historical feedback defects, limited model and instance coverage, and confounding between reference depth and layer count constrain the conclusions. CP2Seq supplies a verifiable planning environment and exposes requirements for more controlled evaluation.

Keywords: origami; spatial reasoning; geometric reasoning; benchmark; multimodal language models; sequential planning

1 INTRODUCTION

A locally valid action can lead away from a target. Completing a folding sequence therefore requires selecting among alternatives, tracking their consequences, and revising earlier decisions. We study long-horizon planning when the model receives the exact coordinates, orientations, stacking order and original-sheet regions of every layer, and every action is checked. Origami provides a generated state space with programmatically checkable success; we use it as a controlled planning task rather than a model of physical folding.

We introduce **CP2Seq**: 600 generated instances, a configurable generator, and an interactive evaluation environment. Given a crease pattern and a target folded state, the model constructs a sequence of simple folds from an unfolded square. Reference sequences and intermediate reference states are withheld; any valid sequence producing both the specified creases and the target stack is accepted. Original-sheet identity and layer order distinguish geometrically overlapping constructions (Section 4.2).

Existing origami benchmarks assess visual questions, pattern prediction or generation (Spencer et al., 2025; Xu et al., 2025; Agarwal et al., 2026). CP2Seq instead tests executable sequence recovery under explicit action constraints. Fold lines are specified in current folded coordinates

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while target creases are given on the original sheet; planning connects these representations through reflections and layer connectivity. Tools expose legality checks, backtracking, optional legal-action enumeration and target feedback. Search baselines use the same candidate actions, allowing us to distinguish task reachability from model success.

Our contributions are (1) a generated dataset and generator spanning reference depths and two simple-fold action models; (2) an interactive pipeline with step-level checks and replay-based terminal scoring, including original-sheet identity; and (3) matched search baselines, assistance and modality ablations, repeated trials, and observable trajectory analysis. These isolate measurable behavior without attributing it to unobserved internal capabilities.

The models solve some short sequences but no medium or hard ablation instance. Yet random legal folding solves 57% of easy and 96% of some-layers episodes within 80 folds, and BFS solves five of ten medium instances. Removing images does not reduce solve rate. Failures rarely reflect reaching the target without stopping; models explore fewer distinct states than random folding and revisit discarded branches. Repeated trials reveal substantial variance, while a second model family solves 14/20 instances in one episode each. We therefore frame CP2Seq as planning over explicit geometric state, not as a validated measure of visual perception or an isolated internal reasoning faculty. Depth and layer count are confounded in the released all-layers corpus, limiting depth-based conclusions.

2 Related work

2.1 Origami benchmarks and generation

GamiBench evaluates visual questions about folded configurations and crease patterns (Spencer et al., 2025). OrigamiSpace covers prediction, spatial relationships and crease-pattern code generation (Xu et al., 2025); CP2Seq instead supplies the pattern and asks for a verified folding sequence. OrigamiBench synthesises flat-foldable patterns and scores query efficiency, image overlap and CLIP similarity, without verifying final layer order (Agarwal et al., 2026). Corigami uses geometric and semantic evaluation for generation (Zahavy et al., 2026), and Learn2Fold plans against a learned world model (Huang et al., 2026). FoldingAgent reconstructs sequences from instructional video (Moriya et al., 2026a), using the human-authored PurelandFold corpus (Moriya et al., 2026b); CP2Seq neither trains nor evaluates on it. Our targets are generated by recorded legal sequences, and final stacks are checked layer by layer and by original-sheet region.

2.2 Computational origami and sequence planning

Simple folds operate along one line at a time; single-, some- and all-layers models restrict which layers move (Arkin et al., 2004). Foldability complexity depends on paper shape, crease directions and permitted operations (Akitaya et al., 2017; 2020; 2023). Layer ordering is central to flat-folded states (Bern & Hayes, 1996; Akitaya et al., 2026; Ku et al., 2026; Eppstein, 2019). Continuous reachability results (Demaine et al., 2004; Demaine & Mitchell, 2001) do not establish reachability by a restricted sequence of simple folds. Sequence recovery from crease patterns is an established computational problem (Akitaya et al., 2013); CP2Seq applies it to language-model planning with executable actions and matched search baselines. Because each target has a legal reference sequence, benchmark reachability is promised, and general foldability hardness does not establish hardness of this generated distribution (Appendix A.5).

3 CP2Seq: task and dataset

3.1 Task and action models

Given a crease pattern C and a target folded state $S^\star$, the model must construct a sequence of legal folds $a_1, \ldots, a_T$ starting from an unfolded square. Replaying the sequence must reproduce the crease pattern and reach the target state under the equivalence defined in Section 4.2. Each state records paper geometry and layer ordering; the reference sequence and its intermediate states are withheld, and alternative valid sequences are accepted. Figure 1 summarizes the task.

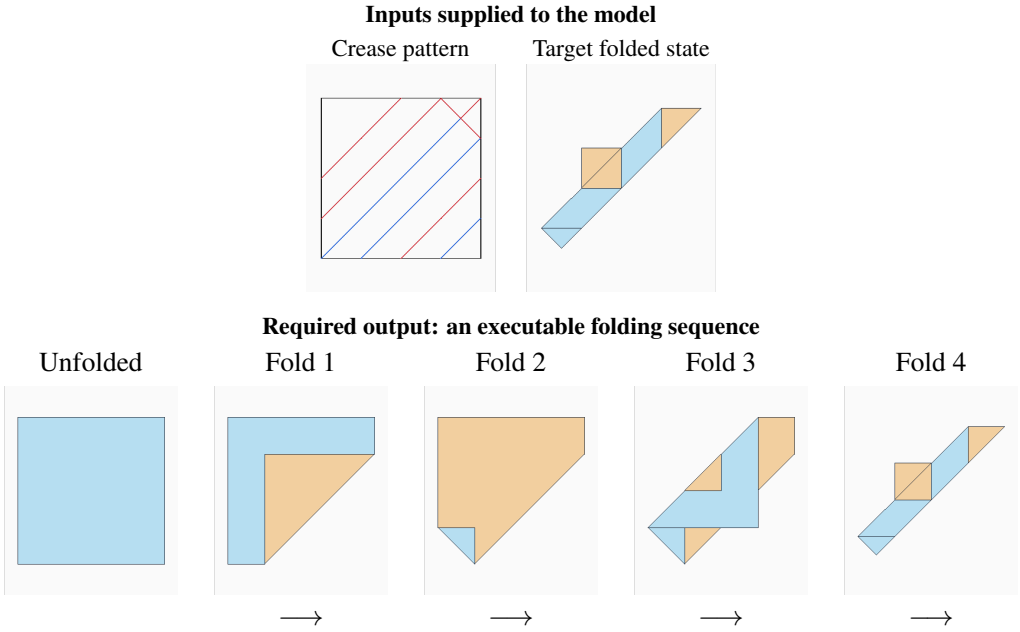


Figure 1: CP2Seq task illustrated by all-layers/easy-0001. The model receives the crease pattern and target state and must produce a valid sequence from the unfolded sheet. The lower row shows the recorded reference sequence, which is withheld during evaluation.

We use two simple-fold action models (Arkin et al., 2004): all-layers folds move the paper on the selected side of a fold line across the full stack, while some-layers folds select a contiguous run from the top or bottom of the stack. Both use flat 180° folds, with connectivity checks to reject moves that would tear the sheet; state representations and action constraints are detailed in Appendix A.

3.2 Dataset construction

Starting from a square sheet, the generator samples a legal fold under the chosen action model, applies it, and records the action and resulting state. This process continues to the requested reference depth; attempts with no available legal continuation are discarded. The accumulated creases are mapped to the original sheet to form the input crease pattern, with edges split at intersections. The last state supplies the target, and the recorded actions supply a reference solution. We group samples by reference fold count, hypothesizing that longer sequences generally require more search and are therefore harder to recover. These groups describe expected difficulty; reference length need not equal the shortest valid solution length. We also record coupling, the number of creases created by a fold, to characterize the geometry of each instance.

Generation pseudocode is provided in Appendix B.3.

We provide inspection software to visualize crease patterns and folded states, replay reference sequences, and examine verification feedback. These views support manual review to identify samples with inconsistent geometry or failed checks for exclusion from the dataset. Example interfaces and rejection feedback appear in Appendix B.

3.3 Dataset statistics

CP2Seq contains 600 samples: 400 all-layers examples grouped by reference depth and 200 some-layers examples with three to six reference folds (Table 1).

The model receives the crease pattern and target folded state; reference actions and intermediate reference states are withheld. Sample contents are documented in Appendix B.7.

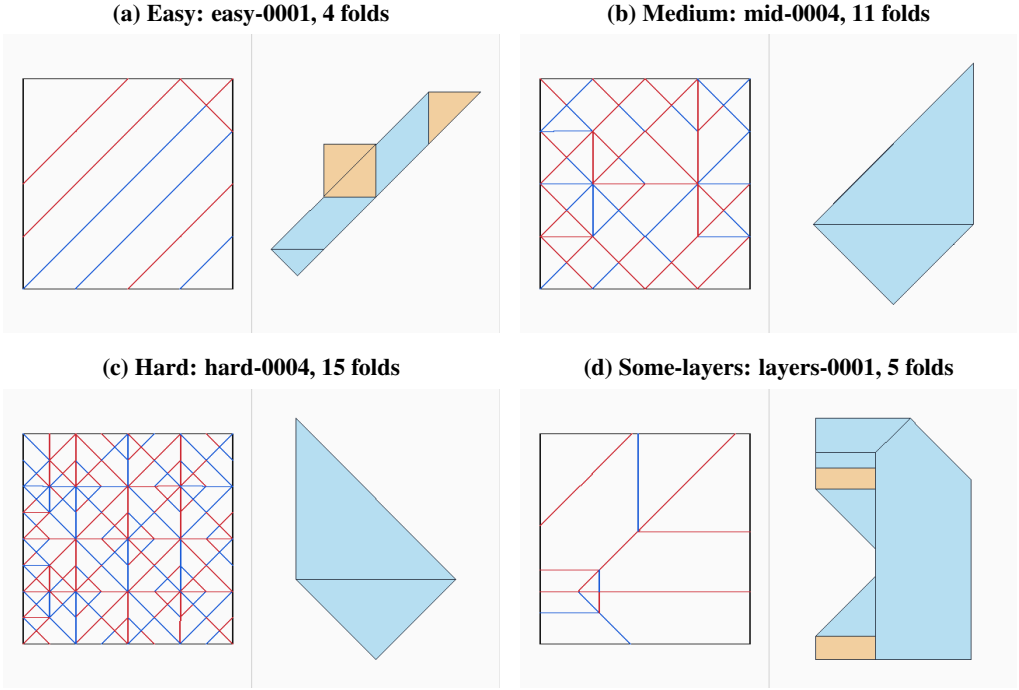


Figure 2: CP2Seq examples: each panel pairs the crease pattern (left) with the target top view (right). Panels (a–c) use all-layers folds; (d) is from some-generated-d5. Counts are reference folds. Red and blue denote mountain and valley creases.

Table 1: Dataset composition. Fold counts refer to recorded reference sequences.

Action model	Group	Samples	Reference folds
All-layers	Easy	200	4–10
All-layers	Medium	100	11–13
All-layers	Hard	100	14–19
Some-layers	Depth 3	50	3
Some-layers	Depth 4	50	4
Some-layers	Depth 5	50	5
Some-layers	Depth 6	50	6
Total		600	3–19

The two replay-based corpus checks pass on all 600 samples; their shared-engine validation scope is discussed in Section 6.

4 Experimental Setup

4.1 Environment

The model edits a folding sequence through the tools in Table 8. For each proposed fold, the engine reflects the selected paper across the fold line, updates layer order and orientation, and checks sheet connectivity and compatibility with the input crease pattern. Accepted edits update the current state; rejected edits return an error code and preserve the previous state.

For a some-layers fold, moving and stationary regions may share an edge away from the hinge. Reflecting only the moving region would separate that shared edge, so the engine rejects the proposal as `would-tear`. A `no-crease` rejection means that paper was selected to move but the operation did not create a crease segment. Other rejections cover an empty moving region, an incompatible over/under direction, and creases absent from the target pattern. That last check is a form of as-

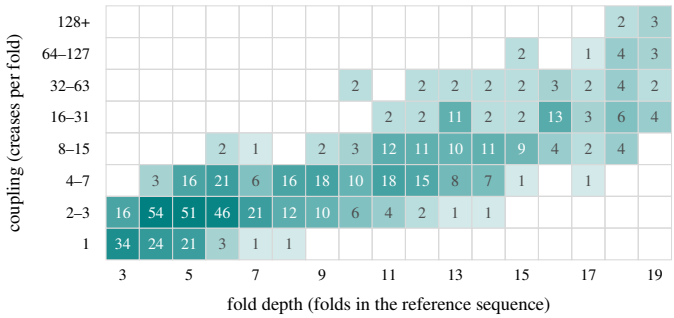


Figure 3: **What the corpus covers, over fold depth and coupling.** Each cell holds the number of samples at that depth and coupling, where coupling is creases made per fold, binned by doublings because an all-layers fold that crosses a doubled stack makes twice the creases. The two axes are not independent: a fold of all layers thickens the stack, so a long sequence cannot keep cutting few layers. No sample reaches 15 folds at coupling below 4, and the deepest such sample has 14; reaching that corner in practice takes pre-creasing, which this action space excludes. The corpus is therefore thinner than real folding in a way the difficulty groups alone do not show.

sistance: it prevents a class of unrecoverable moves, and the rates reported here are therefore rates under that constraint. Detailed conditions are listed in Appendix A.8.

The environment restricts moving layers to the whole stack or a contiguous top/bottom run. It checks flat endpoint states and connectivity, but does not verify continuous collision-free motion; endpoint legality is therefore not a physical realizability guarantee.

Observation tools are listed in Table 8. Exploded views use illustrative layer separation.

4.2 Evaluation metrics

We accept any executable sequence that produces the required crease pattern and folded state.

4.2.1 Verification

We execute the proposed sequence and compare its accumulated creases with the input pattern, including mountain–valley assignments. We then compare the final stack with the target after alignment by a single global translation, rotation, or turnover. A turnover reflects the geometry, reverses the stack order, and flips face orientations. The same transformation must align all layers in their corresponding stack order. A successful construction must satisfy both crease-pattern and folded-state agreement.

Equivalence relation. The submitted results used plane equivalence (E_{plane}): one isometry carries every layer polygon and parity onto the target's at the same rank. The task definition also requires each rank to hold the same region of the original sheet, up to a symmetry of the crease pattern (E_{id}, Appendix A.6). Two stacks equal in the plane but interleaving the sheet differently are different objects. We now score with E_{id}, report both, and count E_{plane}-only solves as unsolved (Appendix A.7).

Audit and scope. Crease-pattern agreement does not remove identity ambiguity. In the reported audit of 536 previously accepted episodes, 514 pass E_{id}; the 22 reversals include 18 model and four BFS solutions. The primary 360-case ablation total changes from 55 to 50 solves (Table 2). Identity-sensitive regression tests include congruent layers with permuted sheet regions; these pass the plane test but must fail the identity test (Appendices A.6 and A.7). These corrections concern terminal outcomes, separately from defects in feedback shown during search.

The crease comparator and the shared folded-state comparator both match positions within 2×10^{-6}; Table 6 lists every tolerance in the pipeline, and Appendix A.6 describes the alignment and representation details.

4.2.2 REPORTED METRICS

We report solve rate by fold-count group, with the number of attempts and the applicable run budgets. Termination reasons distinguish successful submission, unsuccessful completion, budget limits, and repetition-related stopping. Query counts measure calls consumed before termination.

We measure CP edit distance between the accumulated crease pattern in the last recoverable state and the supplied pattern, in original-sheet coordinates. Collinear mountain and valley intervals are merged separately, excluding sheet boundaries. If m intervals match under the crease comparator, the insertion/deletion distance is $d_{\mathrm{CP}} = |C_{\mathrm{target}}| + |C_{\mathrm{candidate}}| - 2m$. Each missing or extra interval costs one; changing an assignment costs two. Partially overlapping intervals do not match unless their endpoints agree within tolerance. We also report $d_{\mathrm{CP}}/|C_{\mathrm{target}}|$, which can exceed one. Normalized distance is undefined for an empty target. Means weight attempts equally and exclude missing state artifacts, whose counts are reported separately.

Verification is operational and tolerance-based, not a formal proof in exact arithmetic. Comparator implementation and validation cases are described in Appendix A.6.

4.3 EVALUATION PROTOCOL

The Luna interface uses OpenAI Codex CLI (OpenAI, 2026) through a custom episode harness; the added Sonnet experiments use Claude Code CLI with the same task tools. Software version evidence is documented in Appendix B.1.

4.3.1 EPISODE PROCEDURE

One episode is one sample attempted by one model under one tier of assistance. The loop is the same for every arm and every model, and the only thing that varies between arms is what the model is allowed to see and call.

Each query contains one tool call. Accepted edits update the current sequence; rejected edits preserve it, and both consume budget. Removing a fold replays the remaining actions; returning to an earlier step truncates the sequence, while revision restoration can recover a previous branch. The reference sequence is withheld until scoring.

Episode pseudocode and the evaluation pipeline appear in Appendix B.4.

All conditions provide fold editing, backtracking, and state observations. Additional conditions provide legal-fold enumeration and target feedback (Section 4.4). The full tool interface and loading configuration appear in Appendix B.5.

4.3.2 MODELS AND BUDGETS

The recent ablations evaluate GPT-5.6 Luna at low reasoning effort and GPT-6 Luna at low and high effort, using ten samples per group and configuration. Runs use an 80-turn limit, a 300-second call timeout, and full image history. Earlier runs have different coverage and budgets and are reported separately (Section 5).

We include breadth-first search over the same enumerated legal actions as a systematic-search baseline (Appendix B.2).

4.4 ASSISTANCE CONDITIONS

The ablations compare tool conditions with common sample IDs and budgets: *basic tools* (B; the interface of Section 4.3.1), *legal enumeration* (L; B plus `list_legal_folds`), and *legal enumeration with automatic feedback* (L+A; L plus tier-3 target feedback after every accepted edit). A fourth condition, B+A, gives automatic feedback without legal enumeration, so that the two additions can be measured separately (Section 5.1). Earlier runs instead let the model request tier-3 comparison on demand.

Automatic feedback supplies layer counts, aligned layer discrepancies, and suggested corrections after edits (Appendix B.5.1). Because those corrections do much of the target comparison for the

model, we interpret feedback conditions as performance with external target-comparison support, not as an isolated measure of the model's own geometric reasoning. Defective feedback can also harm search, so these conditions are not a proven upper bound. Even basic tools provide assistance, since folds that would crease outside the target pattern are rejected (Section 4.1).

5 RESULTS

The main ablations use the first ten fixed-seed instances from each of four groups, with 360 planned sample–configuration pairs. Five protocol failures count as unsolved; retries do not add denominator entries (Appendix B.13). IDs follow generation seeds, not model performance; selected samples match depth but have somewhat fewer layers and creases (Appendix B.15). We report the broader evaluation separately (Table 11), since its samples and budgets differ. All headline solves require CP match and E_{id}; historical E_{plane} counts are explicitly marked. Detailed analyses are in Appendix D.

5.1 TOOL AND REASONING-EFFORT ABLATIONS

Table 2 compares basic tools (B), legal enumeration (L) and enumeration with automatic target feedback (L+A). Pooled totals are 8/120, 15/120 and 27/120. These are descriptive totals over the same 40 instances reused across model settings, not 120 independent tasks. The L+A rows were generated with the defective feedback tool discussed in Section 6; corrected terminal scoring does not make these clean estimates of the effect of correct feedback. We do not pool the earlier, differently budgeted runs into this comparison.

Table 2: Recent Luna ablations. Cells show solved/evaluated samples; each group has ten intended samples. B: basic tools; L: legal enumeration; L+A: legal enumeration with automatic tier-3 target feedback. $*$ includes one input-size error; † includes one service prompt rejection; ‡ includes two omissions after blocking errors. All five count as unsuccessful evaluations. In two L+A cells (easy, GPT-5.6 low and GPT-6 high) one wrong submission followed a false *matches* from the comparison tool (Appendix B.13). Counts are under the identity-aware relation E_{id} (Section 4.2.1). Five easy cells are one lower than under the submitted E_{plane} scoring (marked §; Appendix A.7).

Model	Effort	Tools	Easy	Medium	Hard	Some-layers	Total
GPT-5.6 Luna	Low	B	2/10	0/10	0/10	0/10	2/40
	Low	L	$0/10^{\S}$	0/10	0/10	5/10	5/40
	Low	L+A	$3/10^{\S}$	$0/10^{*}$	0/10	6/10	$9/40^{*}$
GPT-6 Luna	Low	B	0/10	0/10	0/10	0/10	0/40
	Low	L	$0/10^{\S}$	$0/10^{*\dagger}$	0/10	1/10	$1/40^{*\dagger}$
	Low	L+A	$2/10^{\S}$	$0/10^{\ddagger}$	0/10	5/10	$7/40^{\ddagger}$
GPT-6 Luna	High	B	3/10	0/10	0/10	3/10	6/40
	High	L	$2/10^{\S}$	0/10	0/10	7/10	9/40
	High	L+A	3/10	0/10	0/10	8/10	11/40

Feedback without enumeration. A new B+A arm uses GPT-5.6 Luna, low effort, text and images, 80 turns and the same 40 instances. Feedback alone and enumeration alone each raise solves from 2/40 to 5/40; their combination gives 9/40 (Table 3). Single comparisons are underpowered, solved-instance sets differ, and repeated trials vary. We do not infer additive effects or stable per-instance benefits.

Table 3: Separating the two additions (GPT-5.6 Luna, low effort, text and images, E_{id}, one episode per instance; a protocol failure counts as unsolved, so every total is out of 40). B, L and L+A rows are from Table 2; B+A is a new run with the same settings.

Tools	Easy	Medium	Hard	Some-layers	Total
B	2/10	0/10	0/10	0/10	2/40
B+A	3/10	0/10	0/10	2/10	5/40
L	0/10	0/10	0/10	5/10	5/40
L+A	3/10	$0/10^{*}$	0/10	6/10	$9/40^{*}$

Reasoning effort. GPT-6 Luna solves 6/30 some-layers sample–condition pairs at low effort versus 18/30 at high effort. Clustering by the ten instances gives a difference of 0.40 (95% CI 0.23–0.53; exact sign-flip $p = 0.008$; 10,000 resamples, seed 1). These are five-reference-fold instances; the result does not establish performance on long sequences.

Matched search baselines. Table 4 uses the same 40 instances, legal candidate actions and identity-aware success check. Random folding uses 100 independent 80-fold episodes per instance; its per-episode success is 0.572 on easy and 0.959 on some-layers. Model conditions reach at most 5/10 and 8/10 respectively in one episode each. BFS solves 5/10 medium instances within 10^5 expansions. Search establishes that these targets are accessible; its larger total budget does not establish lower cost. Random folding checks success after each accepted fold, whereas model episodes can consume turns on rejections, observations and backtracking and must submit a solution. The comparison therefore matches the legal action set and a nominal 80-step ceiling, not the number of executed folds, stopping rule, inference compute or elapsed time. Only hard is unsolved by every tested method. Its final stacks have a median of 404 layers across the full release, so failure cannot be attributed to depth alone.

Table 4: Matched search baselines on the 40 ablation instances (no model; CP match plus E_{id}). Random: 100 episodes of 80 uniformly random legal folds per instance. BFS: instances solved within the given number of expansions. Seed 1.

	Random, 80 folds		BFS, instances solved within			
Group	instances $\geq$1 solve	per-episode rate	80	10^3	10^4	10^5
Easy	9/10	0.572	7	9	10	10
Medium	3/10	0.005	0	0	3	5
Hard	0/10	0.000	0	0	0	0
Some-layers	10/10	0.959	9	10	10	10

5.2 Input modality

Removing all images while retaining exact numeric state gives 3/40, 5/40 and 11/40 solves with B, L and L+A, versus 2/40, 5/40 and 9/40 with images. With B, images alone give 1/40 (Table 5; Appendix B.16). These small differences do not establish a modality advantage, but provide no evidence that images are necessary. We frame the configured task as planning over explicit geometry, rather than visual perception.

Table 5: Input modality with basic tools (GPT-5.6 Luna, low effort, E_{id}, one episode per instance, totals out of 40). Images only: no coordinates anywhere, only the current stack's bounding box (Appendix B.16).

Input	Easy	Medium	Hard	Some-layers	Total
Text only	3/10	0/10	0/10	0/10	3/40
Images only	1/10	0/10	0/10	0/10	1/40
Text and images	2/10	0/10	0/10	0/10	2/40

5.3 Failure analysis

Cycling terminates 191 of 305 unsuccessful completed ablation episodes, but a stopping condition is not a causal explanation. Trajectory categories are observable mechanical rules (Appendix B.14), not internal capability diagnoses. An automatic stop at the first CP plus E_{id} match raises solves only from 25 to 27 in 143 saved episodes: most failures never reach the target.

The median ratio of distinct model states to random-folding states is 0.11–0.25 for text-only easy and some-layers conditions. Model episodes are often shorter (20–45 turns on average versus 80 random folds), which accounts for part of this gap. In 120 GPT-5.6 Luna L+A episodes, bounded search finds a repair within three removed folds for 71/104 full restarts and 161/280 shorter backtracks. These observations do not establish why a model restarts. After the first backtrack, 340/931 Luna accepted

folds revisit abandoned states versus 26/243 for Sonnet; such revisits need not be errors unless the state has no legal continuation (Appendix D).

Repeated trials and a second family. Five text-only Luna L+A repeats on 20 instances solve 7, 10, 6, 10 and 9 instances (42/100 episodes), with coverage of 14/20 across five trials. Sonnet 5.5 solves 14/20 in one episode each (easy 5/10, some-layers 9/10); one timeout case counts as unsolved, and its first four easy episodes used an earlier prompt order. These small samples and differing repeat budgets do not establish a model ranking.

6 LIMITATIONS

Scope and difficulty. The environment models zero-thickness flat states and simple folds, without material mechanics or continuous collision checks. It is not a physical folding simulator. Reference depth need not be minimum solution depth. In the uncapped all-layers release, depth and layer count are confounded (medium and hard medians 94 and 404 layers; maximum 16,384), so hard-group failure is not an isolated planning-depth limit. A layer-capped corpus was generated, but no reported model result uses it.

Evaluation and uncertainty. Most ablations use one episode per instance and ten instances per group. The repeated Luna trials show substantial variance; five-trial coverage uses five times the budget. Sonnet extends family coverage but does not support broad rankings. Protocol failures count as unsolved. The original comparison tool used E_{plane} and could falsely report a match; identity-aware re-scoring corrects terminal outcomes, not the feedback already shown to models (Appendices A.7 and B.13). For the separate bottom-layer alignment defect, a call-level audit of 65 saved non-ablation episodes finds one divergent call among 1,025 comparisons; two additional calls report a plane-only match that fails identity-aware scoring. Missing image-run logs prevent a complete historical audit. Affected cases have not been rerun, and the direction of the feedback effect remains unknown. An offline stricter cycling threshold can truncate saved trajectories, but a relaxed threshold requires new continuations. We have not run those continuations and do not claim invariance to the stopping rule (Appendix B.14).

Interpretation and reproducibility. Revisits, restarts and stopping labels locate behavior without identifying an internal memory or reasoning deficit. Planned support interventions have not been evaluated. Per-turn logs survive only for the text-only runs; image-run feedback cannot be audited call by call. Cycling limits and unequal episode lengths affect observed exploration. Corpus replay shares the generation engine, numerical checks use stated tolerances, closed model weights cannot be pinned, and training overlap is unknown. Extended limitations and repeated-trial details appear in Appendix E.

7 CONCLUSION AND FUTURE WORK

CP2Seq supplies generated, reachable targets and executable sequence planning with identity-aware verification. Matched search baselines show that model failures on easy, some-layers and medium instances do not imply unreachable targets. Images add no measured benefit over exact numeric state; assistance and effort improve some short-sequence totals. Repeated trials, stopping audits and trajectory analysis characterize observable behavior while exposing variance, comparison-tool defects and depth–layer confounding. Future work should evaluate the layer-capped corpus, repeat every condition, expand model coverage and test controlled support interventions before attributing failure to internal capabilities.

REPRODUCIBILITY STATEMENT

The task, action models and state representation are specified in Section 3.1 and Appendix A; the generation procedure in Section 3.2 and Appendix B.3; the scoring rule, comparator and every numerical tolerance in Section 4.2 and Appendix A.6; the episode loop, tool interface, tool schemas and prompt in Appendices B.4 and B.5; model versions, reasoning effort, turn and timeout limits in

Section 4.3.2; the software version in Appendix B.1; and the accounting of every planned ablation case, including failures, in Appendix B.13. The evaluated models are closed and served through the Codex CLI (Luna) or Claude Code CLI (Sonnet), so their weights and sampling cannot be pinned; we release the saved runs with each attempt's model identifier and reasoning effort, so reported numbers can be audited if not regenerated exactly. The corpus generator, environment and verifier, corpus checks, state comparator and evaluation harness are released under the BSD 3-Clause licence, with code and data available at `https://github.com/Leanmcp-Community/CP2Seq`. Every sample is synthesised by folding a square, so the data carry no personal, human-subject or scraped content; Flat-Folder's example collection is neither used nor redistributed.

USE OF LARGE LANGUAGE MODELS

Language models are evaluated as agents in the folding task. AI assistants were also used for implementation, literature-search assistance, and manuscript drafting and editing. Claude Code was used for coding assistance. AI-generated figure drafts served as composition references; dataset geometry and experimental results must be rendered from the corresponding artifacts. The authors are responsible for the manuscript and released implementation.

DATA AVAILABILITY

The dataset, generation code, and evaluation framework are available at `https://github.com/Leanmcp-Community/CP2Seq`.

CONFLICTS OF INTEREST

Both authors are employed by LeanMCP. The authors declare no other conflicts of interest.

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A Folding model and verification

A.1 Flat-folded states

A state represents the positions of paper regions in the plane, their orientations, and their layer ordering. Original-sheet coordinates identify each region and preserve its connectivity to other regions as folds change its current position. Overlapping regions can occupy the same planar area while remaining distinct layers. Their original-sheet identities therefore matter both when validating a move and when comparing the completed construction with the target. Layer ordering is also a central component of computational treatments of flat-folded states (Akitaya et al., 2026).

A.2 Supported actions

The simple-fold literature distinguishes operations by the layers they move (Arkin et al., 2004). Our all-layers action selects the paper on one side of a fold line throughout the stack. Our some-layers action restricts that selection to a contiguous run from the top or bottom. An action specifies the line in current folded coordinates, the moving side, the layer selection, and whether the moving paper

folds over or under the stationary paper. The selected paper is reflected across the line, with the corresponding orientation and stack-order updates. In the implemented some-layers model, a top run folds over and a bottom run folds under.

A.3 CONNECTIVITY AND TEARING

A move must preserve the connections between regions of the original sheet. If moving paper is attached to stationary paper along a segment away from the proposed hinge, the operation would separate connected paper and is rejected with a `would-tear` response. The check uses original-sheet connectivity together with the proposed motion; overlap in the current plane alone cannot establish whether two regions are attached. Rejection leaves the current state unchanged.

The engine also checks the action parameters and compatibility with the target crease pattern. It represents zero-thickness flat states and does not test continuous collisions during the folding motion.

A.4 DEPTH AND SEARCH DIFFICULTY

Reference depth is the number of generation steps. Search effort also depends on the branching structure and alternative solutions. For random proposal, the probability of selecting a legal move depends on both the legal set and the proposal distribution.

A.5 COMPUTATIONAL COMPLEXITY

The sequence-generation method of Akitaya et al. (2013) and the hardness proofs concern different contributions. For a precise hardness result, Akitaya et al. (2017, Theorem 3) prove that deciding simple foldability of square paper with assigned creases at multiples of 45° is strongly NP-complete in their some-layers and all-layers models. The input is an arbitrary crease pattern; the question is whether any permitted sequence folds it. Strong NP-hardness persists when numerical magnitudes are polynomially bounded.

The reduction starts from 3-PARTITION: divide integers into triples with a common sum. In the orthogonal-paper construction (Theorem 1), pleating a staircase encodes choices of integers. A bar must pass through successive cage slots; the geometry forces each group of choices to contain three integers with the required sum. A partition therefore yields a folding sequence, and a complete folding yields a partition. Theorem 3 embeds this construction in square paper using additional creases that enforce the preparatory folds. The construction has polynomial size, and a proposed folding order can be checked in polynomial time.

For CP2Seq, each target is generated by a recorded legal sequence, so reachability is promised rather than decided. The model must recover a sequence that also matches a supplied terminal state, under the implemented action rules and numerical tolerances. The theorem above does not by itself prove hardness of this promised reconstruction problem or of our sample distribution. Its relevance is that general sequence selection can encode combinatorial choices even when individual actions can be checked efficiently. In an episode, the practical search burden depends on the available actions, solution depth, and the number of unsuccessful branches explored; fold count alone does not determine that burden.

A.6 COMPARISON IMPLEMENTATION

Equivalence relation. Write a flat-folded state as a stack $(\ell_1, \ldots, \ell_n)$, bottom to top, where layer $\ell_i = (P_i, \pi_i, S_i)$ has current-plane polygon P_i, face parity π_i, and original-sheet region S_i (its polygon in the coordinates of the crease pattern). Two final states are *plane-equivalent* (E_{plane}) if some plane isometry T, with turnover acting as above, gives $T(P_i) = P'_i$ and $\pi_i = \pi'_i$ for every rank i. They are *identity-equivalent* (E_{id}) if, in addition, there is a symmetry σ of the square sheet that maps the crease pattern, with its assignments, onto itself (possibly the identity) such that $S_i = \sigma(S'_i)$ for every i under the same alignment. E_{id} is the relation the task definition intends (Appendix A.1): two stacks that look the same in the plane but place different parts of the sheet at some rank are different folded objects. Quotienting by σ is an explicit task convention: a single assignment-preserving symmetry may relabel the entire original sheet, but independent per-layer relabelings are rejected.

The comparator used for the originally submitted results implemented E_{plane}. We now score with E_{id} and report both: an E_{plane} solve that fails E_{id} is counted as unsolved in the primary revised tables. Historical E_{plane} summaries are explicitly identified; the number of reversals is given with each result (Appendix A.7).

The engine compares creases in original-sheet coordinates after grouping and merging collinear intervals by assignment. Interval endpoints are matched within 2×10^{-6}.

The shared terminal comparator first checks layer counts. It considers direct alignment and turnover, with turnover reversing layer order and face parity. Correspondences between vertices of the first layer propose global alignments; a candidate alignment is accepted only if every layer polygon and parity matches its target rank. Polygon matching uses a positional tolerance of 2×10^{-6}, short-edge cleanup at 10^{-7}, and a collinearity threshold of 10^{-9}.

Table 6 collects every numerical tolerance the reported results depend on. Each result record carries the values it was scored under, so a re-scoring can be distinguished from a comparison at the same settings.

Table 6: Numerical tolerances used by the reported checks.

Value	Where used	What it admits
2×10^{-6}	Crease comparator; terminal comparator	Interval endpoint and polygon vertex positions, after an arbitrary global isometry.
10^{-7}	Terminal comparator	Short-edge cleanup before polygon matching.
10^{-9}	Terminal comparator	Collinearity threshold for vertex removal.
2×10^{-6}	CP edit distance	Interval endpoint agreement.
2.44×10^{-4}	CP edit distance	Line quantization when grouping collinear intervals.
10^{-5}	CP edit distance	Gap below which two intervals on one line are merged.

The identity-aware comparator (E_{id}, Section 4.2.1) attaches to each layer its original-sheet polygon, read from the engine state, and requires that, under the alignment found for the plane check, the layer at every rank occupy the same sheet region as the target's layer at that rank, up to one crease-pattern symmetry σ of the square sheet. Sheet regions are compared in the crease pattern's own frame, which both constructions share, so no isometry is applied to them; a turnover reverses the stack and carries each region with its layer. Target sheet regions come from replaying the recorded reference sequence. The symmetry group is computed per sample as the elements of D_4 about the sheet centre that map every mountain and valley segment onto a segment of the same assignment. A layer without sheet data makes the check fail rather than fall back to E_{plane}.

Comparator test cases cover translations, rotations, turnovers, and equivalent polygon representations, including changes in start vertex and winding. Negative cases include incorrect stack order or face orientation, missing layers, scaling, inconsistent per-layer transformations, and empty stacks. Asymmetric polygons distinguish reflections from rotations. Identity-sensitive cases add a stack of three congruent layers whose sheet regions are permuted, which E_{plane} accepts and E_{id} rejects; invariance of E_{id} under rotation, translation and turnover; failure on missing sheet data; and, on replayed reference sequences of `easy-0001`, `easy-0002` and `mid-0001`, acceptance of the reference against itself and rejection of the same stack with its sheet regions cyclically shifted by one rank. All cases pass (`test_terminal_match.mjs`).

A.7 IDENTITY AUDIT

E_{plane} and E_{id} can disagree only when every layer polygon and parity already agrees. We measured how often this happens in two ways.

Accepted solutions. We replayed all 536 episodes scored as solved in the supplementary runs and compared each with the replayed reference under both relations. All 536 replay legally and pass E_{plane}; 514 pass E_{id}. None of the 22 reversals is explained by a crease-pattern symmetry. Eighteen are model solutions and four are breadth-first search solutions, whose goal test used the same plane-only comparator. All 22 are on easy samples, and they concentrate on few patterns: 14 of the 18 model reversals are on `easy-0003` (10 folds) and `easy-0004` (5 folds), the rest on `easy-0053`, `easy-0088` (twice) and `easy-0127`. In `easy-0004`, for example, the two stacks agree on 32 congruent layers in the plane, but 28 of the 32 ranks hold a different region of the sheet: the construction folds the same flaps in a different interleaving. No medium, hard or some-layers solve is affected.

Five of the reversals fall in Luna ablation cells, one each in GPT-5.6 Luna L and A, GPT-6 Luna low L and A, and GPT-6 Luna high L, all on the easy group; Table 2 reports the corrected counts (55 to 50 of 360 solves). The four breadth-first reversals are in the supplementary search runs (928 attempts); we could not match them to the 725-attempt snapshot of Appendix B.2, so we state the bound: at most four of its 366 easy solves fail E_{id}.

Adversarial rollouts. For each of the 40 ablation samples we drew 200 random legal rollouts of exactly the reference depth from the flat sheet, sampling uniformly among the distinct legal folds the model's own enumerator returns, and checked every terminal state. Of the 8,000 rollouts, 3,226 reach the reference depth without a dead end; 1,536 of these reproduce the crease pattern, 1,210 satisfy E_{plane}, and 1,202 satisfy E_{id}. The 8 that satisfy E_{plane} but not E_{id} are all on easy instances: seven on `easy-0007` and one on `easy-0003`. On the other 38 instances the two relations never disagree, including 977 E_{plane}-equivalent rollouts on the ten some-layers instances. No medium or hard rollout satisfies E_{plane}. Plane-equivalent but identity-distinct constructions are therefore a property of a few specific patterns, which a model or a search can reach by chance. The audit script and its outputs are `workspace/audit_identity.mjs` and `workspace/RESULTS/identity-audit/`.

A.8 Action rejection conditions

Table 7: Geometric and action constraints checked by the implementation. Rejected edits preserve the previous state.

Code	Condition
`would-tear`	A moving region and a stationary region share an original-sheet edge whose current position is not on the fold line. Reflecting one side would separate the join.
`no-crease`	A nonempty moving selection produces no crease segment in the paper. This condition is checked separately from tearing.
`nothing-to-move`	The selected half-plane contains no selected paper, or the requested partial selection contains no layers.
`direction-impossible`	The action requests folding a top run under the stack or a bottom run over it, contrary to the supported layer-selection rule.

The independent Flat-Folder wrapper uses layer-order constraints, including taco-taco, taco-tortilla, tortilla-tortilla, and transitivity constraints (Akitaya et al., 2026; Ku, 2022). It tests whether the pattern admits a flat-folded state.

B Software and supplementary examples

B.1 Software versions

The inspected Codex CLI installation is version `0.155.1` (OpenAI, 2026). Its upstream release tag `rust-v0.155.1` resolves to commit `be2951ea34f0d295ed0becf97079f92fa5f6950e`.

The saved run configurations record the executable path but do not record its version or binary hash; the installation was checked on 25 September 2026.

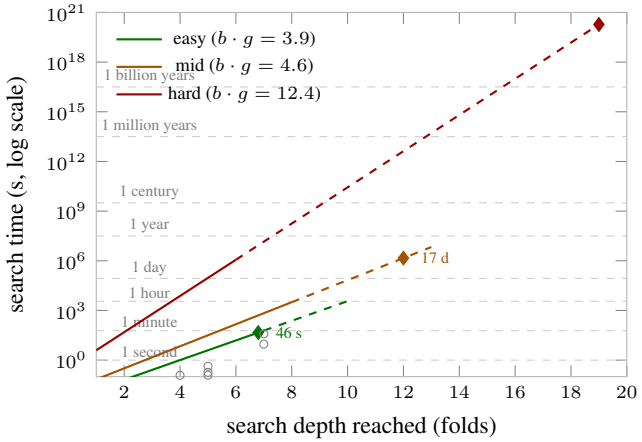


Figure 4: **Breadth-first search time against search depth, by difficulty group.** The vertical axis is logarithmic, so a straight line is exponential growth and its slope is $\log(b \cdot g)$. Curves are $t(d) = c_0(b \cdot g)^d$, the product of two exponentials: the frontier grows as b^d for a deduplicated branching factor b, measured as generated over expanded during search, while the cost of expanding one node grows as g^d, because one enumeration evaluates $4 \times$ (distinct crease lines) $\times$ (facets) candidates and a fold of all layers at most doubles the facets. g and c_0 are fitted by least squares on per-depth enumeration timings. Each curve is solid only where that group was observed and dashed beyond, and spans only the depths its own corpus reaches. Diamonds are each group's mean reference depth; the hard group's sits at 10^{20} seconds, which is why the axis runs that far. Circles are observed solves. b is fitted from budget-exhausted runs: 7 over 3 samples for easy, 12 over 3 for medium, and 8 over 2 for hard, across 5 time budgets. The hard extrapolation rests on two samples. Extrapolated values are model projections, not measured solution times or lower bounds.

B.2 Breadth-first search analysis

The saved BFS evaluation contains 725 attempts: 366/669 solved on easy samples (54.7%), 1/40 on medium, and 0/16 on hard. These attempts include repeated samples and different budgets and are not paired with the recent ten-sample ablations. BFS uses wall-clock, state-expansion, and depth limits, while model runs use turn and call-timeout limits.

A fold is specified by a line in current-plane coordinates together with a layer selection, and both are drawn from small finite sets. Candidate lines are read off the crease pattern itself: the distinct lines through its mountain and valley edges, carried forward through each distinct face transform into current coordinates, then deduplicated. On a box-pleated pattern thousands of crease edges collapse to a few dozen lines. Layer selections are not arbitrary subsets either, because a partial fold must be a contiguous run at one extremity of the stack; the selections are the whole stack, the top k, and the bottom k, which is about $2N$ for N layers rather than 2^N, with the over or under direction forced for partials. Each surviving candidate is judged by the same function the `add_fold` tool calls, so a listed fold cannot be one the tool would then reject, and it is dropped if it fails any material check of Section 4.1 or would crease the sheet where the target pattern carries no crease.

Across 73,750 enumerations recorded in the saved runs, this evaluates a mean of about 2,010 candidate actions per state and returns 3.51 of them, a ratio of approximately 570 between evaluated and returned candidates. The baseline expands with the same enumerator the model is given, tests goals with the same terminal comparator that decides `solved`, reads no reference sequence, and runs under a per-sample wall-clock budget with caps on states expanded and depth.

The supplementary cost model approximates search time as the product of two exponential factors: the frontier grows with the branching factor, and the cost of expanding one node grows with depth as well, because one enumeration evaluates four candidates per crease line per facet and a fold of all layers at most doubles the facets. Fitting both from the saved searches gives a per-fold multiplier of 3.9 on the easy group, 4.6 on medium and 12.4 on hard (Figure 4). These fitted curves extrapolate beyond the observed depths; they are not measured solution times or lower bounds. The hard-group

Group (reference folds)	Median states expanded	Solved / attempts
Easy (4–10)	65	366/669
Medium (11–13)	1,051.5	1/40
Hard (14–19)	532	0/16

Figure 5: Observed BFS search effort in the saved aggregate. Bar lengths use a linear scale and show median states expanded across attempts, including unsuccessful attempts. Budgets and repeated samples are not standardized by this aggregate; 358 of 725 attempts terminated in timeout.

fit uses only two samples, so its factor is illustrative only; because depth is confounded with layer and crease count across groups, none of the fitted factors should be read as the intrinsic complexity of the task. No matched, fixed-budget comparison of search and models on the same instances has been run.

B.3 Dataset generation

```
GENERATE(reference_depth, action_model):
    state <- unfolded square
    actions, frames <- [], []
    for step in 1..reference_depth:
        candidates <- LEGAL_FOLDS(state, action_model)
        if candidates is empty: return FAILED
        action <- SAMPLE(candidates)
        state <- APPLY(state, action)
        actions.append(action)
        frames.append(state)
    cp <- CREASE_PATTERN_IN_ORIGINAL_SHEET(state)
    cp <- SPLIT_EDGES_AT_INTERSECTIONS(cp)
    if not REPLAY_CHECK(cp, actions, state): return FAILED
    return (cp, state, actions, frames, METADATA(actions))
```

B.4 Evaluation pipeline

```
EPISODE(sample, model, condition, query_budget):
    cp, target <- LOAD_INPUTS(sample)
    session <- INITIALIZE(cp, target, unfolded_square)
    history <- []
    for query in 1..query_budget:
        obs <- OBSERVE(session, condition)
        call <- model(cp, target, obs, history, condition)
        if call is finish: break
        # Edits: add/remove fold, go to step, restore revision.
        # Reads: state, images, candidates, target feedback.
        result <- DISPATCH_TRANSACTIONALLY(session, call)
        # Rejected edits leave session unchanged but consume a query.
        # Read calls consume a query without changing the sequence.
        history.append(call, result)
        if STOPPING_RULE(session, history): break
    return SCORE(REPLAY(session.actions), REFERENCE_AT_SCORING(sample))
```

B.5 Tool interface and feedback

Basic tools comprise all entries in Table 8 except the optional `list_legal_folds` and `compare_to_target` tools. Loading and configuration are described in Appendix B.10.

B.5.1 Target-feedback levels

The run configuration selects one of three feedback levels for `compare_to_target`. The tool compares the current and target stacks.

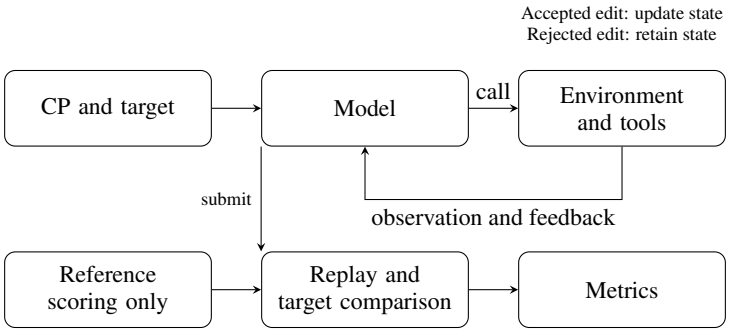


Figure 6: Evaluation pipeline. The model edits and inspects its current construction. The reference sequence is withheld from the model and used only for scoring.

Table 8: Tools available to the evaluation harness; exposure depends on run configuration.

Tool	Operation
`add_fold`	Append a checked fold; reject invalid actions without changing state.
`remove_fold`	Remove a specified fold and replay the remainder transactionally.
`go_to_step`	Retain the first specified number of folds.
`restore_revision`	Restore a previously recorded branch state.
`get_state`	Return the current sequence and layer geometry.
`get_images`	Render top, oblique, reverse, X-ray, and exploded views of a completed step; exposed views depend on run configuration.
`list_legal_folds`	Optionally enumerate legal candidates compatible with the target CP.
`compare_to_target`	Optionally return feedback at the configured level (Section 4.4).
`finish`	End the episode and evaluate the current sequence.

B.6 TOOL SCHEMAS AND PROMPT

The following listings document the tool schemas and folding prompt; tool exposure is configured per run.

```
"""Schemas supplied to the native Qwen3.5 tool-aware vision renderer."""
import json


def tool(name, description, properties=None, required=None):
    return {"type": "function", "function": {"name": name, "description": description,
            "parameters": {"type": "object", "properties": properties or {},
                           "required": required or [], "additionalProperties": False}}}


TOOLS = [
    tool("add_fold", "Append a 180-degree fold. Choose angle_index OR angle_degrees. Partial
     ↪ top/bottom runs must not tear connected paper.", {
        "angle_index": {"type": "integer", "enum": [0, 1, 2, 3]},
        "angle_degrees": {"type": "number", "description": "Line angle counterclockwise from +
     ↪ x; offset uses unit normal (-sin(theta), cos(theta))."},
        "selection_mode": {"type": "string", "enum": ["all", "top", "bottom"]},
        "layer_count": {"type": "integer", "minimum": 1, "description": "Required for top/
     ↪ bottom, omit for all."},
        "offset": {"type": "number"}, "move_positive": {"type": "boolean"},
        "over": {"type": "boolean"}}, ["offset", "move_positive", "over"]),
    tool("remove_fold", "Remove 1-based fold; replay the rest transactionally.",
         {"step": {"type": "integer", "minimum": 1}}, ["step"]),
    tool("go_to_step", "Keep the first N folds; zero clears the sequence.",
         {"step": {"type": "integer", "minimum": 0}}, ["step"]),
    tool("restore_revision", "Restore a returned revision, including abandoned branches.",
         {"revision": {"type": "integer", "minimum": 0}}, ["revision"]),
    tool("get_images", "Render top, two oblique, X-ray, and exploded-stack PNGs at a completed
     ↪  step.",
         {"step": {"type": "integer", "minimum": 0}}),
    tool("get_state", "Inspect the current sequence and bottom-to-top layer polygons."),
```

Table 9: Information returned at each target-feedback level.

Level	Information returned
1: Counts	Current and target layer counts, their difference, and a count-based summary.
2: Diagnosis	Level 1 plus aligned layer matches, mismatched ranks, and a diagnosis of count, shape, orientation, or ordering disagreement.
3: Corrections	Level 2 plus suggested corrections for up to twelve mismatched ranks, such as orientation changes or a different target rank for a layer shape.

```
    tool("list_legal_folds", "List the folds that are legal in the current state and on-target
     ↪  for the CP. "
         "Read-only. Every listed action is accepted verbatim by add_fold while the state is
     ↪ unchanged.", {
        "max_results": {"type": "integer", "minimum": 1},
        "selection_filter": {"type": "string", "enum": ["any", "all", "top", "bottom"]},
        "include_rejected": {"type": "boolean"}}),
    tool("compare_to_target", "Compare the current stack with the target folded state. Read-
     ↪ only. "
         "Detail is fixed by the run's configured tier, not by you. Uses the target state only
     ↪ ; "
         "the reference fold sequence is never consulted."),
    tool("finish", "End this episode and evaluate the current sequence."),
]

# The Codex loop exposes only a subset in its baseline condition; the enumerator changes
# what the benchmark measures, so it must be requested explicitly. See TODO.md.
ENUMERATION_TOOLS = ("list_legal_folds",)
COMPARE_TOOLS = ("compare_to_target",)


def _whole_stack_only(tools):
    """Strip partial folds from the schemas, so nothing offers what ToolSession will refuse.

    Two edits, and both are needed. add_fold loses selection_mode and layer_count, or the
    model can request a run the simulator then rejects; list_legal_folds loses
    selection_filter, whose remaining values would all be lies once the run forces "all".
    """
    out = []
    for t in tools:
        name = t["function"]["name"]
        if name not in ("add_fold", "list_legal_folds"):
            out.append(t)
            continue
        copy = json.loads(json.dumps(t))
        props = copy["function"]["parameters"]["properties"]
        for key in ("selection_mode", "layer_count", "selection_filter"):
            props.pop(key, None)
        if name == "add_fold":
            copy["function"]["description"] = (
                "Append a 180-degree fold of the whole stack. Choose angle_index OR
     ↪ angle_degrees.")
        out.append(copy)
    return out


def tools_for(mode, compare_tier=0, action_space="any"):
    """The action set for one experimental condition.

    compare_to_target appears only when a tier is configured, so the default arms are
    byte-identical to every run recorded before the comparison tool existed. action_space
    defaults to "any" for the same reason: nothing changes unless a run asks for it.

    action_space is a property of the corpus, not a preference. On release/all-layers every
    one of the 4180 reference folds is whole-stack, so partial folds cost 3-87x in search and
    buy nothing; on some-verified-d3 and some-generated-d6 they are what 14% and 34% of the
    samples require. Whichever is chosen must be given to the search arm too, or the
    comparison measures the setting rather than the solver.
    """
    if mode == "base":
        chosen = [t for t in TOOLS if t["function"]["name"] not in ENUMERATION_TOOLS]
    elif mode == "legal-folds":
        chosen = list(TOOLS)
    else:
        raise ValueError(f"Unknown tool mode: {mode}")
    if not compare_tier:
```

```
        chosen = [t for t in chosen if t["function"]["name"] not in COMPARE_TOOLS]
    if action_space == "all-layers":
        chosen = _whole_stack_only(chosen)
    elif action_space != "any":
        raise ValueError(f"Unknown action space: {action_space}")
    return chosen


Recover a folding sequence from the supplied crease pattern (CP) and final folded
state. The sequence starts empty. Use one tool call at a time, then inspect its
result before proposing the next fold. Do not return an entire sequence in prose.
The environment writes your accepted actions into seq.json.

You receive only geometry and images of the CP, final target, and current state.
No reference folding sequence is supplied. All folds are flat 180-degree simple
folds. Select all layers (default), or a contiguous run from the top or bottom.

Coordinates are x right, y up. Fold axes are in CURRENT folded coordinates:

| angle_index | fold line | positive side |
| --- | --- | --- |
| 0 | y = offset | y > offset |
| 1 | y = x + offset | y - x > offset |
| 2 | x = offset | x > offset |
| 3 | x + y = offset | x + y > offset |

For arbitrary line angles, supply angle_degrees instead of angle_index, never
both. With theta counterclockwise from +x, the line is
-sin(theta)*x + cos(theta)*y = offset; offset is signed perpendicular distance.
move_positive selects the greater-than side. The folding motion remains 180 degrees.
selection_mode is all (default), top, or bottom. Top/bottom requires layer_count;
omit layer_count for all. Top must fold over; bottom must fold under.
The engine tracks original-sheet connectivity. A partial fold returns would-tear
if selected paper is attached to stationary paper away from the proposed hinge.
For example, selecting three of four layers may tear a paired flap. This check
runs before target-CP checking and leaves state unchanged on rejection. Inspect
sheet_polygon coordinates and exploded views when choosing a connected run.

move_positive selects which half-plane moves. over=true puts the reflected moving
stack above the stationary stack; false puts it below. A flipped stack reverses
its internal order. Offset is not a normalized distance for diagonal axes.
The tool rejects extra creases or incompatible mountain/valley assignments and
leaves the state unchanged on error. This is a zero-thickness geometric engine;
it does not check continuous collisions during motion.

add_fold appends ONE fold. remove_fold removes a 1-based fold and replays the
remaining actions; an invalid replay leaves the sequence unchanged. go_to_step
keeps the first N folds (0 empties the sequence). restore_revision restores a
previous tool state, even after changing branches; revision 0 is the empty sheet.
These editing operations create new revision IDs. get_state returns your current
sequence and geometry. get_images returns PNG views of any completed step in the
current branch without changing it. Use the returned revision IDs when backtracking.

Images: CP red = mountain, blue = valley, dark boundary = paper edge. Top and
two oblique views show paper shape with a small illustrative layer separation.
Oblique views look down from opposite directions, never exactly edge-on. X-ray
is a top projection: more overlapping layers produce darker pixels. Each view
is fitted independently; use geometry for coordinates and scale, not pixel size.
Exploded views separate layers upward for inspection: layer 1 is the bottom,
and p0/p1 labels give face parity. Separation is illustrative, not paper thickness.
Initial images include the target's exploded view; feedback images include the
current state's exploded view after edits.

Try to reproduce both the CP and the supplied target, including face orientation
and layer order. Terminal matching accepts any global translation and any rotation
angle, including 30 degrees. It also accepts turning the whole model upside down:
reflect the geometry, reverse the bottom-to-top stack, and flip every face parity.
These operations can be combined. One common rigid transformation must match all
layers; arbitrary layer rearrangements, scaling, or independently moving layers
do not count. CP matching still uses the supplied original-sheet coordinates and
mountain/valley assignments. Call finish when you are done. This ends the episode and
    ↪ evaluates your
current sequence; it does not automatically claim success.
```

B.7 SAMPLE CONTENTS

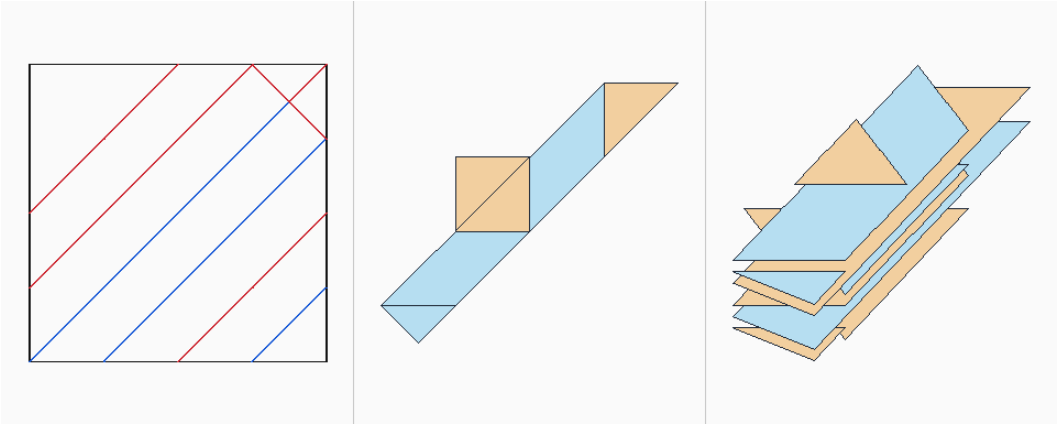
Each sample contains the input crease pattern, a reference state sequence, reference fold actions, and metadata (Table 10).

Table 10: Contents of a CP2Seq sample.

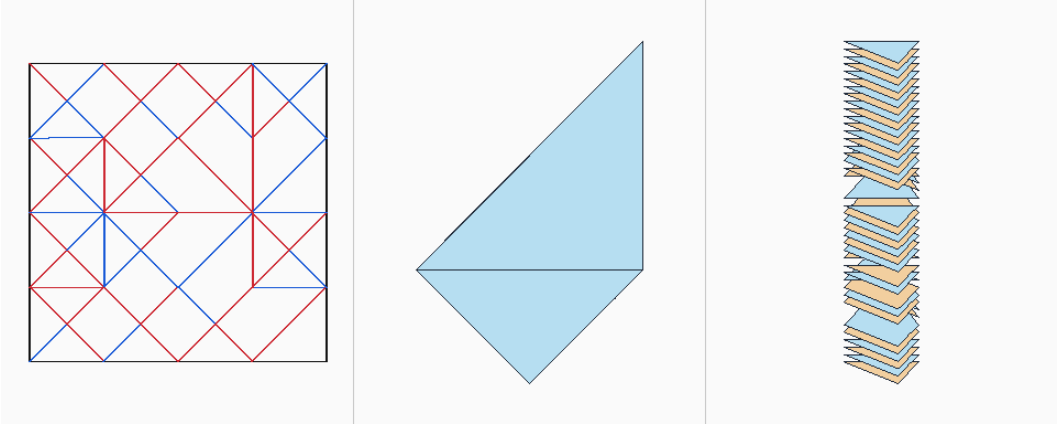
Artifact	Contents
Crease pattern	Input crease geometry.
State sequence	Folded states after each reference action; only the final state is supplied as the target.
Reference actions	Withheld fold lines, directions, layer selections, and creases created.
Metadata	Sample descriptors and generator version.

B.8 Dataset gallery and reference sequences

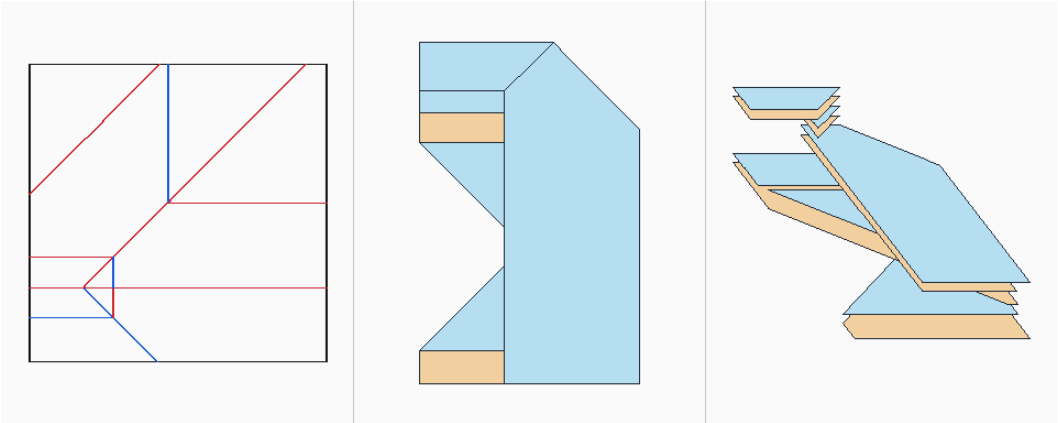
Each rendered panel contains the crease pattern, top view, and an oblique view with illustrative layer separation. all-layers/easy-0001 (all-layers, 4 reference folds)



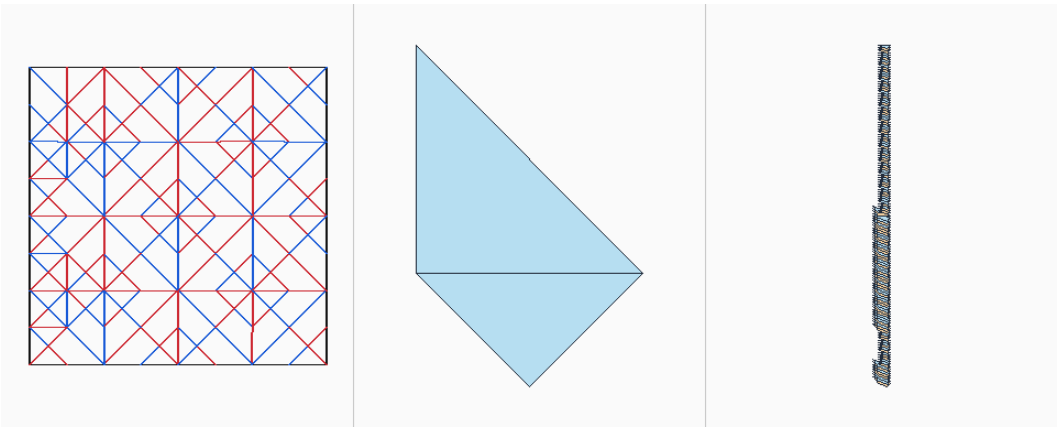
all-layers/mid-0004 (all-layers, 11 reference folds)



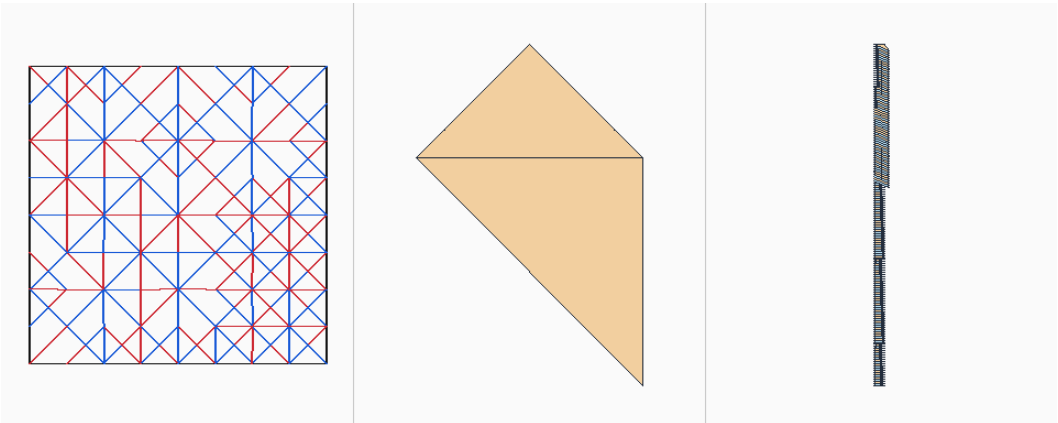
some-generated-d5/layers-0001 (some-layers, 5 reference folds)



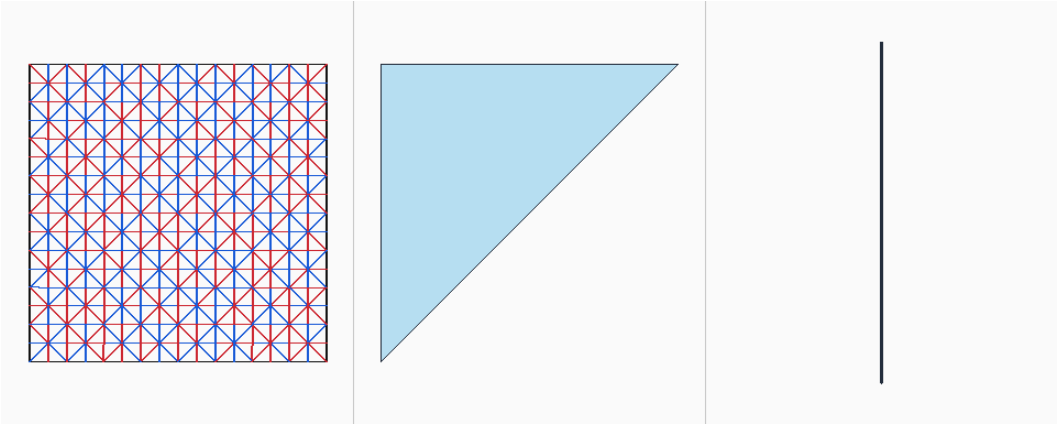
all-layers/hard-0004 (all-layers, 15 reference folds)



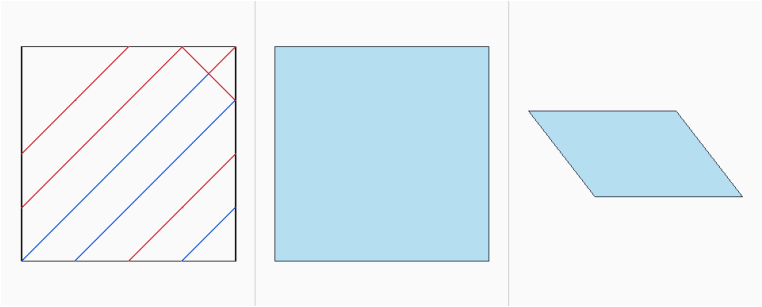
all-layers/hard-0005 (all-layers, 15 reference folds)



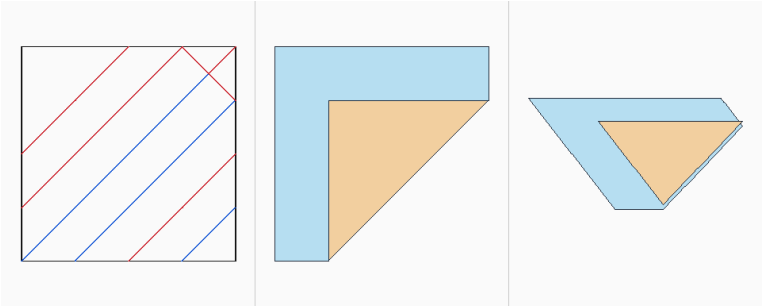
all-layers/hard-0006 (all-layers, 15 reference folds)



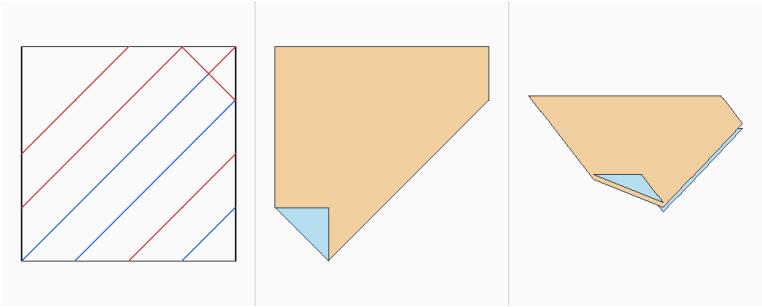
all-layers/easy-0001 (all-layers, 4 reference folds) Stored state 0



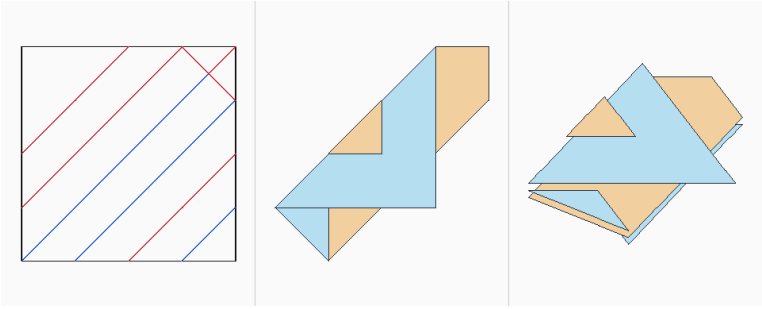
Stored state 1



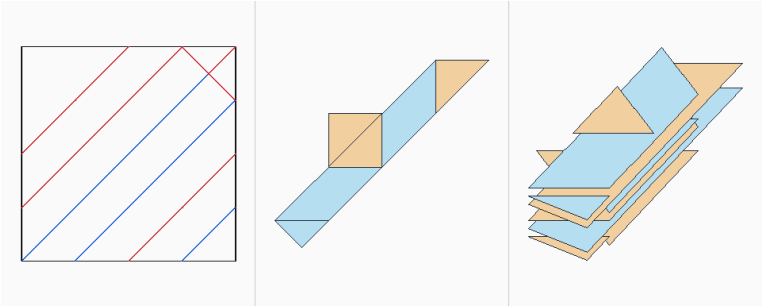
Stored state 2



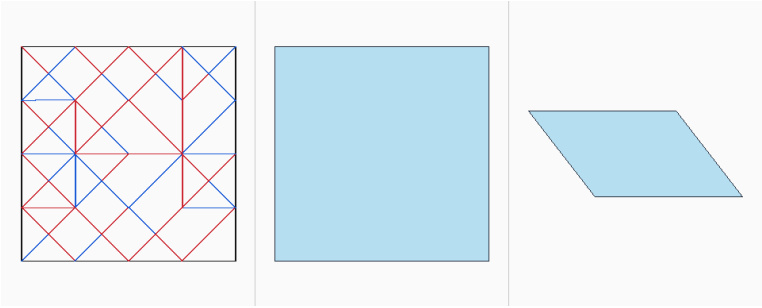
Stored state 3



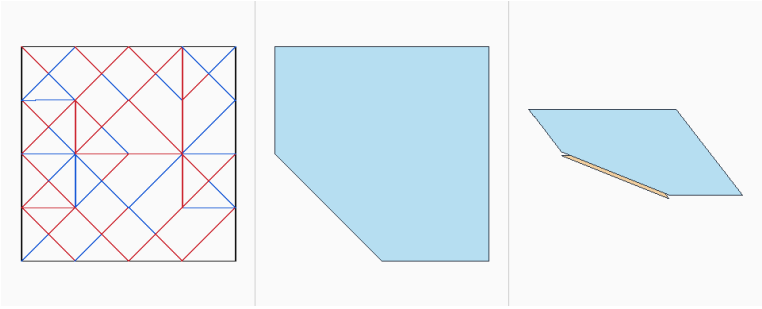
Stored state 4



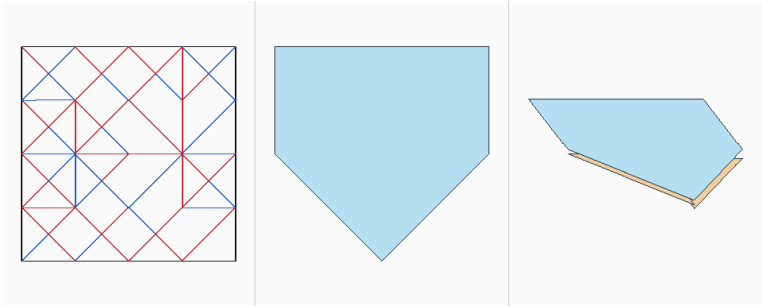
all-layers/mid-0004 (all-layers, 11 reference folds) Stored state 0



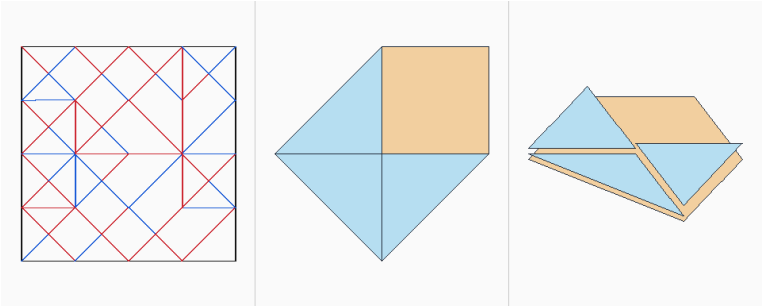
Stored state 1



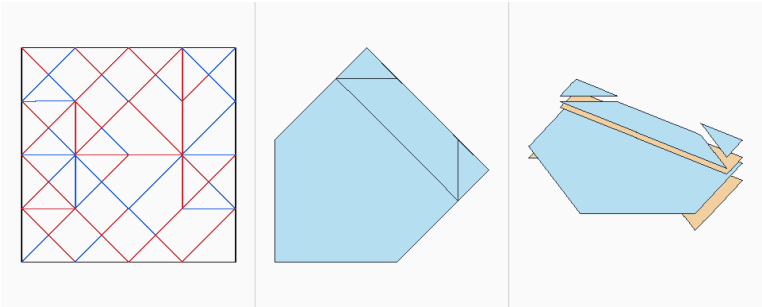
Stored state 2



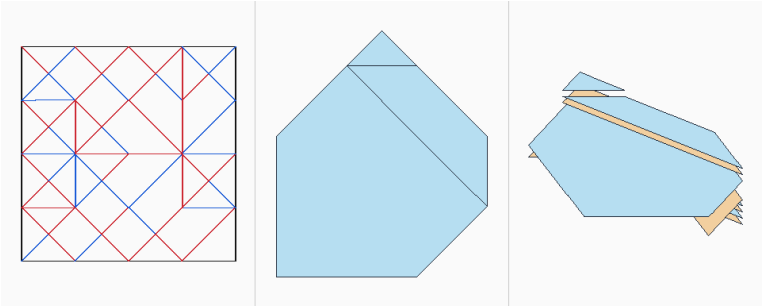
Stored state 3



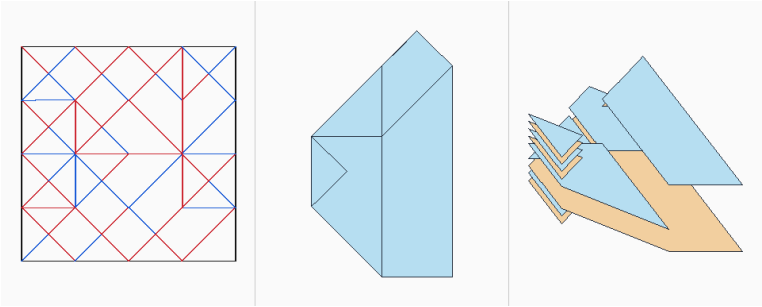
Stored state 4



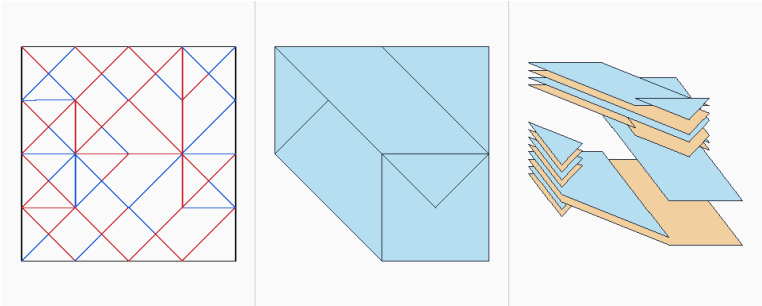
Stored state 5



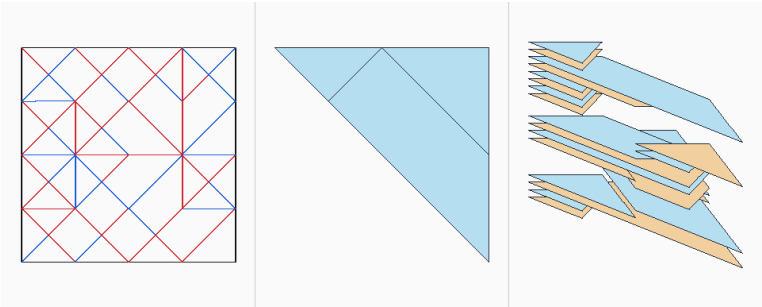
Stored state 6



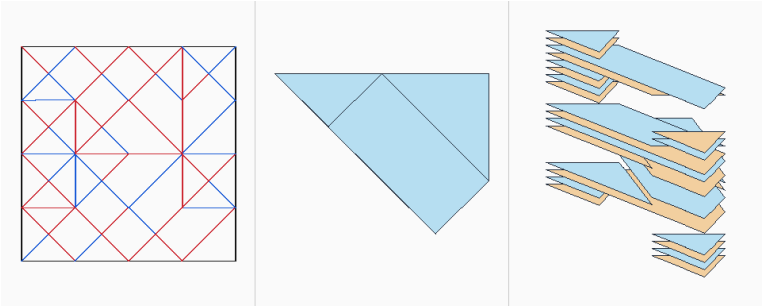
Stored state 7



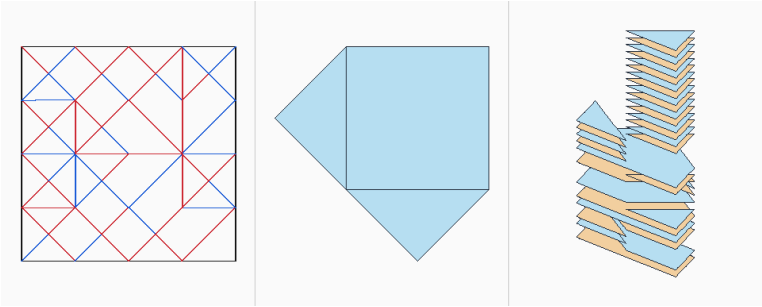
Stored state 8



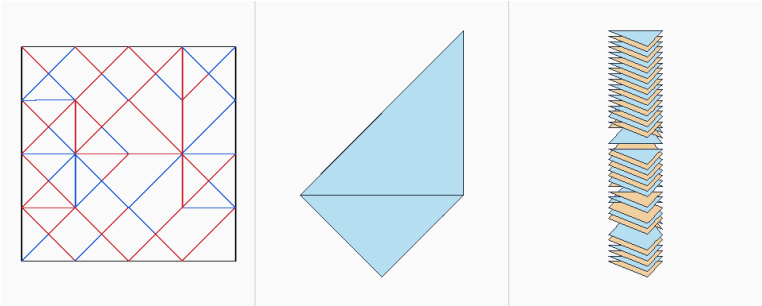
Stored state 9



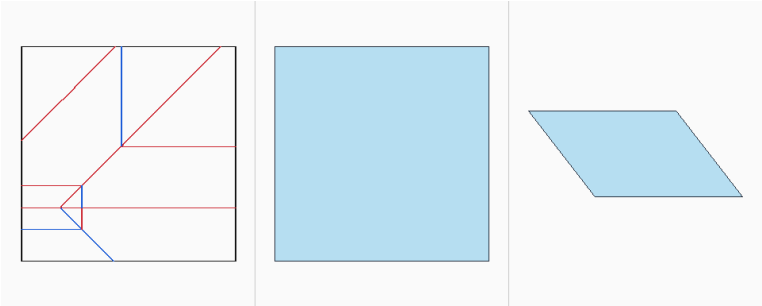
Stored state 10



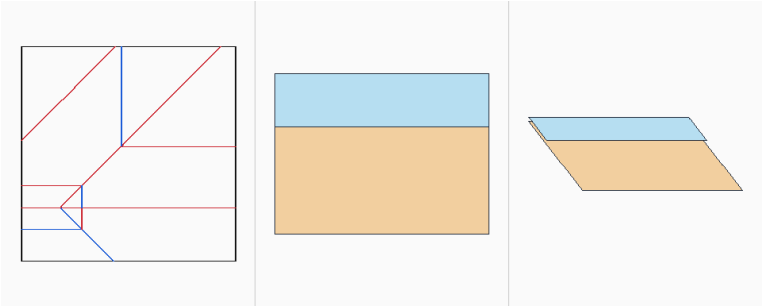
Stored state 11



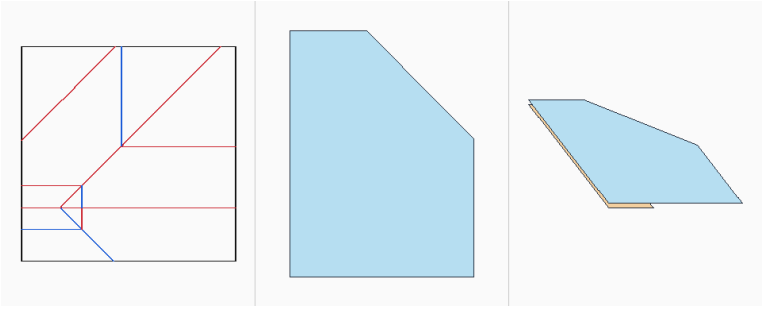
some-generated-d5/layers-0001 (some-layers, 5 reference folds) Stored state 0



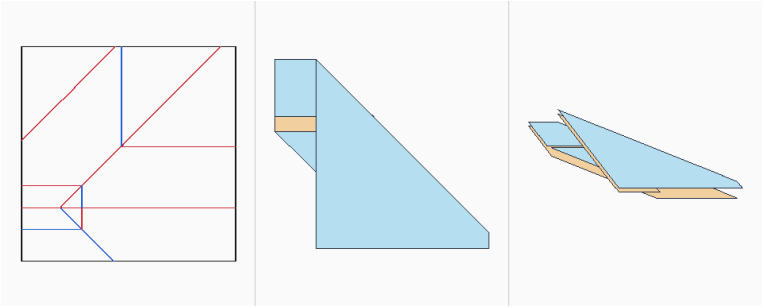
Stored state 1



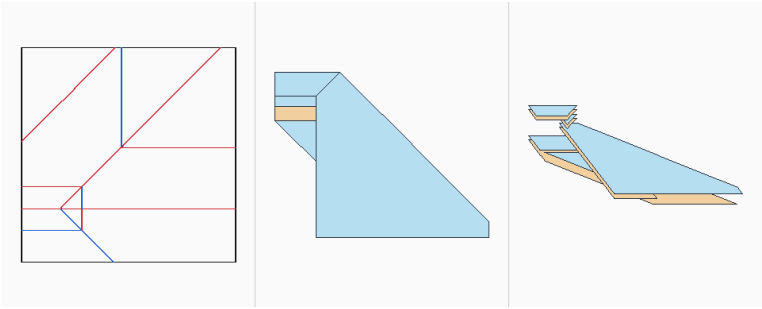
Stored state 2



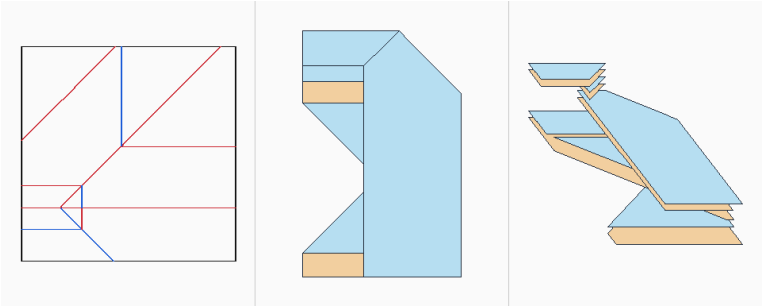
Stored state 3



Stored state 4



Stored state 5



B.9 TRACE-ANALYSIS INTERFACE

We provide a trace-inspection interface for browsing saved runs, replaying recorded folding trajectories, and examining model actions alongside tool feedback and rendered states. This supports qualitative analysis of successful construction, repeated exploration, and unsuccessful termination. A command-line trace viewer additionally supports filtering by task, trial, session, role, and event type.

Figures 7 and 8 show the dataset browser and trace viewer, respectively. These interfaces support inspection of reference sequences and recorded model episodes.

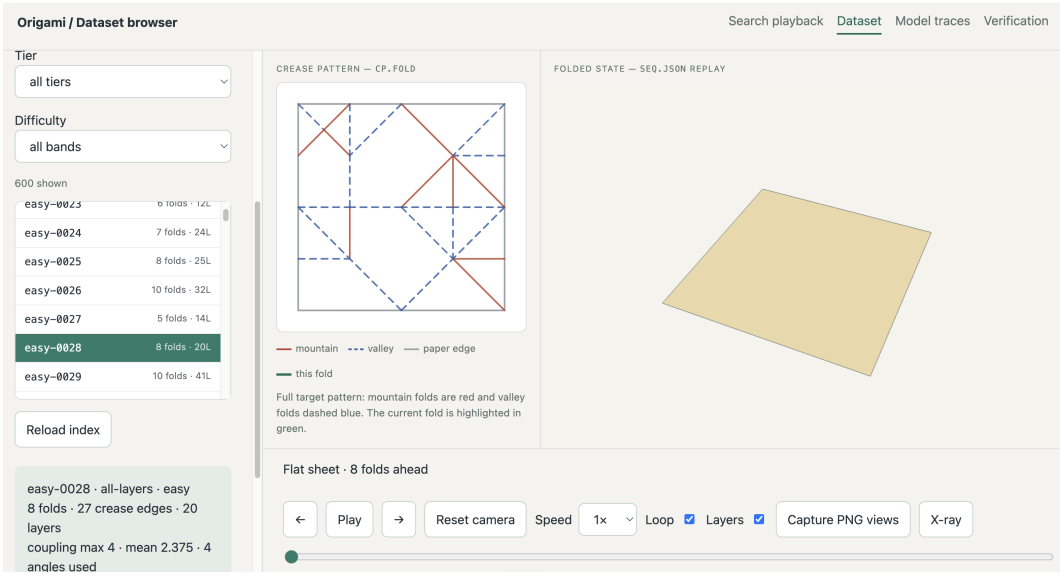


Figure 7: Dataset browser with `easy-0028` selected. The interface displays sample metadata, the target crease pattern, and reference-sequence playback controls. The right panel shows the unfolded sheet at step zero.

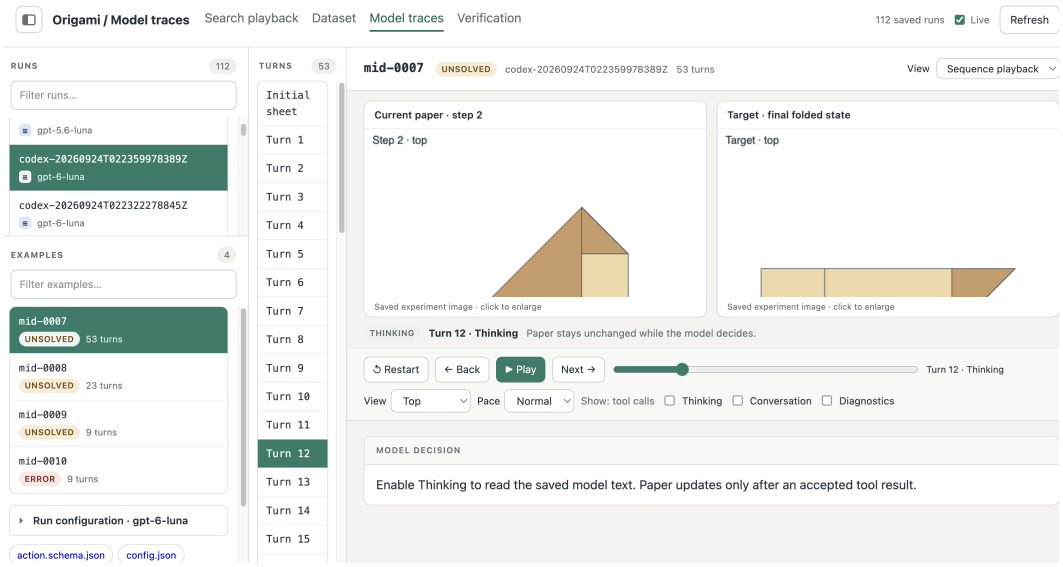
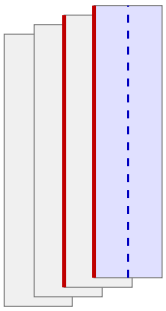


Figure 8: Trace viewer for a saved GPT-6 Luna attempt on `mid-0007`. Run and turn navigation appear beside the current and target states. The selected episode is marked unsolved; the displayed frame is at turn 12. Playback controls allow inspection of saved actions and feedback.

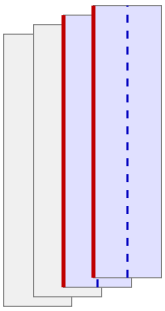
(a) Rejected proposal



`would-tear`

Top leaf selected; state unchanged.

(b) Accepted proposal



`ok: true`

Connected top pair selected.

Figure 9: Rejection feedback and a legal alternative from the same state. Dashed blue lines mark the proposed hinge; red segments mark the shared material edge identified by the tearing diagnostic. Blue shading identifies selected layers. Selecting only the top leaf would separate it from its connected neighbour away from the hinge; selecting the connected top pair permits the fold. Both panels show the pre-action state, with illustrative layer separation. The example is rendered from verification fixtures.

B.10 Tool loading and configuration

The schema builder selects the tools exposed in each run. Basic mode excludes candidate enumeration; filtered mode includes it. Target comparison is added only when a comparison level is configured. For an all-layers action model, partial-layer selection arguments are removed from the exposed action schemas. The run configuration therefore records tool mode, comparison level, and action model separately.

B.11 Rejection feedback

Figure 9 illustrates the geometric evidence associated with a `would-tear` response using the shared verification fixtures. The comparison holds the initial state and hinge fixed and changes the selected layers. The export script records both engine responses and checks that rejection preserves the state.

B.12 Supplementary results

Table 11: Earlier GPT-5.6 Luna evaluation at low reasoning effort. Counts are solved attempts over completed attempts, including repeats across samples and budgets.

Condition	Solved / completed	Solve rate
Basic tools	4/67	6.0%
Legal enumeration	31/272	11.4%
Legal enumeration + on-demand comparison	28/108	25.9%

The inventory also contains small exploratory runs: GPT-6 Sol with legal enumeration solves 1/3 low-effort attempts and 1/1 high-effort attempt; GPT-6 Astra with basic tools solves 1/1 low-effort attempt. Earlier basic-tool runs record 0/7 for GPT-5.6 Sol and 0/1 for GPT-5.6 Terra. One additional completed basic-tool attempt has no recorded model identity and is unsolved. These exploratory runs are reported separately from the Luna ablations.

A separate scan of 1,421 saved attempts with recoverable states gives a mean crease-pattern insertion/deletion distance of 45.5, or 0.55 of the target crease count (Table 14). These averages pool models, baselines, configurations and repeated samples, and predate the ablations.

Table 12: How each Luna ablation episode ended, under E_{id}. Solved: correct submission. Wrong: incorrect submission, including the five submissions accepted under E_{plane} only (Appendix A.7), which moves one episode from Solved to Wrong in five cells. Cycling: stopped after revisiting earlier states. Dead end: no recoverable legal continuation. The five protocol failures of Appendix B.13 have no completed episode and are not listed, so rows can sum to fewer than 40.

Model	Effort	Tools	Solved	Wrong	Cycling	Repetition	Turn budget	Dead end
GPT-5.6 Luna	Low	B	2	6	14	3	5	10
	Low	L	5	13	22	0	0	0
	Low	L+A	9	3	27	0	0	0
GPT-6 Luna	Low	B	0	2	12	20	1	5
	Low	L	1	11	25	1	0	0
	Low	L+A	7	1	29	0	0	1
GPT-6 Luna	High	B	6	4	13	5	5	7
	High	L	9	9	21	0	1	0
	High	L+A	11	1	28	0	0	0
Total			50	50	191	29	12	23

Table 13: Easy-group solve rate by reference fold count, over all completed model attempts in the inventory. Attempts include repeats and differing budgets, so rates are descriptive.

Reference folds	4	5	6	7	8	9	10
Solved / attempts	19/59	35/115	12/46	10/97	3/60	0/33	7/47
Solve rate	32.2%	30.4%	26.1%	10.3%	5.0%	0.0%	14.9%

These attempt-weighted averages pool saved configurations and repeated samples; they are descriptive and do not constitute matched model comparisons. The CP-distance scan may cover more runs than the earlier solve-rate snapshot.

B.13 Protocol failure accounting

All 360 planned Luna ablation cases enter the solve-rate denominator. Of these, 355 have completed episodes. Two cases end at the request input-size limit: `mid-0006` for GPT-5.6 Luna low with automatic feedback and GPT-6 Luna low with legal enumeration. GPT-6 Luna low with legal enumeration on `mid-0010` ends in a service prompt rejection. For GPT-6 Luna low with automatic feedback, `mid-0006` and `mid-0010` were omitted from the restarted batch after these blocking errors. These two cases are classified as omissions after blocking errors; no direct service error is recorded for them in that condition. All five cases count as unsuccessful evaluations under the planned protocol.

Comparison-tool defect. The target-comparison tool aligns stacks by their bottom layer. When no rigid motion aligned the two bottom layers and the layer counts were equal, it reported the diagnosis *matches* with zero matching layers. Scoring uses a separate comparator and was not affected by this defect. Replaying the final state of every incorrect submission made under a comparison condition shows that five followed this false report: `easy-0001` in the automatic-feedback ablations for GPT-5.6 Luna (low) and GPT-6 Luna (high), and `easy-0001`, `easy-0058` and `easy-0095` in the earlier on-demand comparison runs.

Inspecting final states does not bound the defect's effect: a wrong diagnosis at any turn can redirect the rest of the search, and the correction need not only add solves. A full bound requires every comparison the model was shown, and the per-turn logs of the image ablations were not retained, so we cannot audit them call by call or rerun them under identical conditions. The two affected ablation cells (easy, GPT-5.6 Luna low A and GPT-6 Luna high A) should therefore be read as measured under a defective tool, with an effect of unknown sign.

Where per-turn logs survive, in 65 GPT-5.6 Luna episodes with automatic feedback outside the ablations, we replayed the accepted sequence recorded with each of the 1,025 comparisons shown and recomputed it with the corrected tool. One call, in one episode, differs. The comparison tool, like the submitted scorer, used E_{plane}, and two calls reported *matches* on a state that is plane- but not

Table 14: CP insertion/deletion distance over saved attempts. Means exclude unscored attempts. Normalization divides each distance by its target merged crease count before averaging.

Model	Attempts	Scored	Mean distance	Mean normalized
claude-sonnet-5	5	5	12.800	0.481
codex-cli-chatgpt	1	1	8.000	1.000
deterministic-bfs	725	725	38.810	0.494
deterministic-reverse	203	203	24.212	0.764
gpt-5.6-luna	447	447	67.159	0.538
gpt-5.6-sol	7	7	15.857	0.586
gpt-5.6-terra	1	1	0.000	0.000
gpt-6-astra	1	1	0.000	0.000
unknown	32	31	43.065	0.804
All attempts	1422	1421	45.454	0.553

identity-equivalent to the target. On this sample the defect is rare, but the sample is not the ablation set.

B.14 Trajectory analysis

Failure categories. Each unsuccessful episode is assigned the first of four mechanical categories that applies. *F1, state tracking*: more than half of the proposed folds are geometrically illegal in the current state (tearing, no crease, nothing to move, impossible direction, invalid selection). *F2, off pattern*: most rejections are legal folds outside the target crease pattern, and the crease pattern is never completed. *F3, wrong stack*: the episode reaches a state with the full crease pattern and the target layer count, but in the wrong order, orientation or shape. *F4, shallow search*: none of the above; the episode never reaches the target depth with a consistent crease pattern. The categories are descriptive. They separate where in the task an episode broke down, not which internal capability failed.

Table 15: Trajectory-level analysis of the completed text-only GPT-5.6 Luna (low effort) episodes, the runs whose per-turn logs were retained. Rejection: share of proposed folds rejected (all causes / geometric only). Backtracks: accepted remove, go-to-step and restore calls per episode. F1–F4: unsuccessful episodes by category. Outcomes as submitted (E_{plane}).

Tools	Episodes	Solved	Rejection	Geom.	Backtracks	F1	F2	F3	F4
B	39	3	82.0%	20.5%	1.2	3	29	0	4
L	39	7	13.0%	8.4%	3.4	0	5	6	21

Cycling limit. The cycling rule stops an episode when the same fold is played from the same position three times. A looser limit cannot be evaluated from logs, because episodes were not observed past their stop, and we have not rerun any configuration with a different limit. Cycling counts are therefore a property of the protocol as well as of the models.

B.15 Ablation sample selection

Sample selection. Sample IDs are assigned in generation order, which follows consecutive generator seeds (`easy-0001–0010` use seeds 20260918–20260927); the corpus manifest was written on 17 September 2026, before any ablation run (from 23 September), so the selection could not depend on model performance. The selected samples match their groups in reference depth (mean 6.2 vs. 6.8 folds on easy, 11.9 vs. 12.0 on medium, 16.4 vs. 16.2 on hard, 5 vs. 5 on some-layers) but are somewhat simpler in final layer count (26.4 vs. 33.0 on easy, 116 vs. 134 on medium, 860 vs. 933 on hard, 10.5 vs. 11.6 on some-layers) and in crease count. If anything, solve rates on the selected samples overstate those on the full groups.

B.16 Images-only condition

The complementary images-only condition removes every coordinate from the prompt and from tool results: the crease pattern, the target and the current state are given only as rendered images, and

tool results report acceptance or rejection but not the resulting geometry. The renderings are scaled to fit the canvas and carry no axes, so without any numeric cue a model could not state a fold line's offset, which is a coordinate in the current folded frame. We therefore keep one cue: the bounding box of the current stack in current folded coordinates (four numbers), which lets a model convert image positions to coordinates. The prompt states that the sheet is the unit square with x right and y up. This condition uses basic tools only, since legal-fold listings carry coordinates. GPT-5.6 Luna at low effort, 80 turns, one episode per instance on the 40 ablation instances, solves 1/40 under E_{id}: one easy instance, and no medium, hard or some-layers instance (Table 5). The same model solves 3/40 with text only and 2/40 with text and images under basic tools.

C Extended related work

This appendix collects the literature the benchmark rests on, grouped by the role each work plays rather than alphabetically. Section 2 discusses the works the design depends on directly; the rest are recorded here.

Origami benchmarks and systems for language models. Spencer et al. (2025) releases both code and data, and evaluates viewpoint consistency and the recognition of infeasible patterns. Xu et al. (2025) evaluates pattern prediction, multi-step spatial reasoning and crease-pattern code generation, and ships an interactive environment for the last of these. Agarwal et al. (2026) synthesises flat-foldable patterns in an interactive environment over 366 designs collected from Flat-Folder, and evaluates query efficiency, geometric similarity through intersection over union, and semantic similarity through CLIP embeddings; its query efficiency is the proportion of fold steps that contribute to the final model, not the number of queries to a solution. It is one source of the metrics surveyed while our scoring was designed. Zahavy et al. (2026) co-designs flat-foldable and recognisable models, using a vision-language model for aesthetic and semantic evaluation and reporting the flat-foldability of generated crease patterns separately. Huang et al. (2026) plans structured generation against a learned world model. Moriya et al. (2026a) reconstructs folding processes from instructional video, over the sequence-level corpus of Moriya et al. (2026b).

Complexity and theory of flat folding. Bern & Hayes (1996) carries the claim the task rests on: geometry does not determine layer order, and deciding the overlap order of a flat folding is NP hard even when a valid mountain-valley assignment is given. Arkin et al. (2004) defines simple foldability and the one-layer, some-layers and all-layers models this action space is drawn from, and Akitaya et al. (2017) settles the complexity of deciding it for square paper with creases at multiples of $45°$. Akitaya et al. (2020) treats the all-layers model, which is also the model of sheet-metal bending, and Akitaya et al. (2023) treats mixed orthogonal crease patterns. Akitaya et al. (2026) gives a method for deciding whether a pattern admits a flat-folded state at all, a problem that is NP hard in general. Ku et al. (2026) gives an algebraic treatment of the layer-ordering constraints, and Eppstein (2019) studies the graphs of flat foldings themselves. Demaine et al. (2004) and its predecessor Demaine & Mitchell (2001) separate a folded state from a continuous folding motion, for simple polygonal and rectangular paper respectively. Those results concern continuous motion; they do not establish that a folded state is reachable by the restricted sequence of simple folds used here, and the discrete step structure is the question this benchmark poses. Hull & Zakharevich (2023) establishes that flat origami is Turing complete, which bounds what any general claim about foldability can promise.

The crease-pattern-to-sequence problem itself. Akitaya et al. (2013) named the problem and supplied the tools the area still uses: reflection paths, graph rewriting, and a step sequence graph whose paths yield folding sequences. xkevio (2026) is an open-source implementation of that line and is usable as a symbolic baseline, which we have not yet run.

Counting folded states. The number of ways to fold a strip of n labelled stamps runs 1, 2, 6, 16, 50, 144, 462, 1392, 4536, 14060 for $n = 1, \ldots, 10$ (OEIS Foundation Inc., 2026). The problem was posed by Lucas (1891) in 1891 and has been attacked repeatedly since (Koehler, 1968; Lunnon, 1971); the related enumeration of meanders is surveyed by Legendre (2013), which reports that no closed formula was known. The relevance here is that the number of valid states grows fast enough that scoring against any single reference sequence would be wrong, which is why Section 4.2 scores by execution instead.

Spatial reasoning context. Liu et al. (2025) surveys spatial reasoning in multimodal models. Wu et al. (2024) names the mental-imagery question this benchmark turns into a measurement: whether a model can maintain and update a spatial representation across steps rather than describe one.

Statistical physics of random flat foldability. Nakajima (2024) represents layer-order constraints as spin interactions, and Hull et al. (2025) proves that for random locally flat-foldable mountain-valley assignments on the square grid, global flat foldability is exponentially unlikely under its sampling distribution. Neither result concerns the number of terminal states of the patterns in this corpus, and we do not extrapolate them to it.

Tools, formats and data. Ku (2022) is the implementation that decides and enumerates flat-folded states, for the method of Akitaya et al. (2026); together they are the source of the four constraint classes cited in Section 4.1. Demaine et al. (2026) is the de facto standard used by every artifact we ship. Mitani (2007) is a crease-pattern editor with folded-form estimation. Moriya et al. (2026b) is the closest existing sequence-level corpus, with 27 sequences and 337 annotated keyframes released under CC-BY-4.0 as the ground truth for FoldingAgent; it is not used here as training or evaluation data.

Scope, and what is left out. Thick folding (Ku & Demaine, 2016), bar-and-hinge mechanics (Filipov et al., 2017) and its simulators (Zhu & Filipov, 2021; Zhu, 2026), and the wider creased-sheet mechanics literature belong to a different line of work, one in which the sheet has thickness and the verifier is a simulation rather than a decision. That line asks a question this benchmark does not: the states here are zero-thickness and flat, and the environment rules on a fold in one pass rather than integrating a deformation. Those works are therefore not treated as neighbouring work, and the limits of the model used here are stated in Section 6.

D EXTENDED EXPERIMENTAL ANALYSIS

We report the completed Codex runs in the inventory generated on 24 September 2026, separating the broader GPT-5.6 Luna evaluation from the recent ablations. The inventory counts completed attempts, including unsuccessful submissions and stopping-limit terminations. For the ablations, all 360 planned sample–configuration pairs enter the denominator. Execution errors and omissions after blocking errors count as unsuccessful evaluations, with causes marked separately. We retain the first completed attempt per sample and configuration when available; retries do not add denominator entries.

Earlier GPT-5.6 Luna runs used different samples and budgets per condition, so we list them without comparison (Table 11); tool effects are measured only in the ablations.

D.1 TOOL AND REASONING-EFFORT ABLATIONS

The recent study uses the first ten samples of each group in every configuration: `easy-`, `mid-` and `hard-0001–0010` from the all-layers corpus and `layers-0001–0010` from the some-layers corpus; the some-layers samples all have five reference folds. It compares GPT-5.6 Luna at low effort with GPT-6 Luna at low and high effort. Tool conditions and run settings are given in Sections 4.4 and 4.3.2. Of the 360 planned sample–configuration pairs, five are protocol failures counted as unsolved (Appendix B.13).

Sample selection. IDs follow consecutive generator seeds and were fixed before any ablation run; the selected samples match their groups in depth but have somewhat fewer layers and creases, so their solve rates if anything overstate the full groups' (Appendix B.15).

Tools. Summing the Total column of Table 2 over the three model settings gives, for the same 40 samples in each, 8/120 solves with B, 15/120 with L and 27/120 with L+A (two L and three L+A pairs are protocol failures counted as unsolved); the order B<L<L+A holds in each setting. By group, easy gives 5, 2 and 8 of 30 and some-layers 3, 13 and 19 of 30; medium and hard give none. These are descriptive comparisons over reused instances. In particular, the feedback rows were generated under a defective comparison tool, whose trajectory-level effect has unknown sign.

Terminal re-scoring does not remove that confound; we do not interpret these totals as a controlled estimate of the benefit of corrected feedback.

Feedback without enumeration. Because L+A adds two things to B, the ablation above cannot say whether target feedback helps without legal enumeration. We therefore ran B+A (basic tools plus automatic tier-3 feedback, no `list_legal_folds`) for GPT-5.6 Luna at low effort on the same 40 instances, with the image ablation's settings (text and images, 80 turns, one episode per instance; Table 3). All five B+A solves also satisfy E_{id} on replay. Against B (2/40), feedback alone gives 5/40, the same as legal enumeration alone (5/40), and the two together give 9/40. With ten instances per group none of these differences is significant on its own. The totals do not decompose by instance: B+A solves easy-0001, 0004 and 0006, and at most one of these is among the L+A easy solves; L solves layers-0007, which L+A does not. With one episode per instance, which instances a condition solves varies between runs: five further repeats of the text-only L+A run solved 2, 3, 1, 3 and 3 easy instances under E_{id}, against 5 in the original run, with different instance IDs (Section 6). We therefore read Table 3 only as showing that each addition raises the total, not that their effects add or which instances each helps.

Reasoning effort. Pooled over the three tool conditions, GPT-6 Luna solves 6/30 some-layers samples at low effort and 18/30 at high effort. Per condition the increases are 0/10 to 3/10, 1/10 to 7/10 and 5/10 to 8/10; with ten samples per cell only the second is significant on its own (two-sided Fisher exact test, $p \approx 0.02$; $p \approx 0.21$ and 0.35 for the others), and a 95% Wilson interval spans roughly ± 0.3 (e.g. 0.11–0.60 for 3/10). Pairing outcomes by sample and condition, every discordant pair favours high effort (12 to 0; exact McNemar test, $p < 0.001$), so the effect is consistent even where single cells are underpowered. McNemar treats the 30 sample–condition pairs as independent, but each some-layers instance appears in all three tool conditions. Treating the instance as the unit, a cluster bootstrap over the ten instances gives a high-minus-low difference of 0.40 in some-layers solve rate (95% CI 0.23–0.53), and an exact sign-flip test over per-instance differences gives $p = 0.008$. This instance-clustered result concerns the ten five-fold some-layers instances and the observed historical tool conditions; it is not a clean estimate under corrected feedback. The effect magnitude is uncertain and does not establish performance at larger depths.

Depth. No Luna ablation solves a medium or hard sample, and the breadth-first baseline solves 1/40 medium and 0/16 hard attempts (Appendix B.2), against 366/669 on the easy group (at most four of which fail E_{id}; Appendix A.7). Within the easy group, the model solve rate falls from 32% at four folds to 0–15% at eight to ten (Table 13). The groups differ in more than depth. In the all-layers tier every fold moves the whole stack and usually crosses every layer, so the layer count grows roughly exponentially with depth: over the 40 ablation instances the final stack has a median of 23 layers on easy (maximum 64), 90 on medium (448) and 512 on hard (3,504), and over the full released groups the hard median is 404 (maximum 16,384). Depth and layer count are therefore confounded, and the hard group mostly measures states of hundreds to thousands of layers, whose serialised geometry approaches or exceeds what fits in a prompt (one medium episode failed on input size; Appendix B.13). These results show that the deeper groups are harder, not that depth alone is the cause, and we do not read the hard group as a limit of planning depth (Section 6). Breadth-first search time on the saved searches grows by a fitted factor of 3.9 per fold on easy and 4.6 on medium (Figure 4); the hard-group factor of 12.4 rests on two samples and an extrapolation, and we do not use it as a measure of task complexity.

Matched search baselines. The comparison above mixes budgets and instance sets. To compare the model with search on the same footing, we ran two baselines that use no model on the same 40 ablation instances, with the action set the model is given and success defined as CP match plus E_{id}, checked after every fold (Table 4). *Random* plays 80 uniformly random legal folds per episode (the model's turn budget), 100 episodes per instance. *BFS* searches distinct states (identity-keyed) up to the reference depth plus four, and we record whether it reaches the target within 80, 10^3, 10^4 and 10^5 expansions (600 s wall clock per instance). Within an 80-fold budget, random legal folding solves 57% of easy and 96% of some-layers episodes, while the best model conditions solve at most 5 of 10 easy instances (GPT-5.6 Luna, text-only L+A) and 8 of 10 some-layers instances (GPT-6 Luna, high effort, L+A) in one episode each (Table 2, Section 5.2). BFS solves 5 of 10 medium instances within 10^5 expansions; random folding solves at least one episode on 3 of 10 medium instances.

Only the hard group is unsolved by every method we ran, and its states are the largest (above). We therefore do not describe medium as beyond reach: on easy, some-layers and medium the model is less effective than undirected search over the same legal actions. Search with 10^5 expansions and 100 random episodes per instance uses far more attempts than one model episode; the comparison shows that the model's choices carry less information than chance would over the same number of folds, not that search is cheaper or better in general.

D.2 Input modality

To measure what the images contribute, we reran the GPT-5.6 Luna (low effort) ablation with every image removed from the prompt and from tool results, keeping the JSON geometry of the crease pattern, target and current state. Under E_{id}, text-only solves 3, 5 and 11 of 40 with B, L and L+A, against 2, 5 and 9 with images (pooled 19/120 vs. 16/120); under the submitted E_{plane} scoring the text-only counts were 3, 7 and 11 (21/120 vs. 18/120; discordant pairs 9 vs. 6, exact McNemar $p = 0.61$). Removing images does not reduce solve rate on these samples. The prompt already states every vertex coordinate, parity and layer rank exactly, while a 512-pixel rendering shows the same geometry less precisely and cannot show which layer lies above which; the model's successful reasoning runs through the coordinates. At this sample size we cannot exclude a small benefit from images, but we find none. CP2Seq as configured here therefore measures geometric reasoning over an exact numeric description of every layer (coordinates, orientation, stacking order and sheet region) more than visual perception. One text-only basic and two text-only medium episodes ended in protocol failures and count as unsolved.

An images-only condition, which keeps the renderings and removes the coordinates, isolates perception (Appendix B.16). With basic tools it solves 1/40 (one easy instance), against 3/40 text-only and 2/40 with both (Table 5). The differences are one and two instances out of 40 and are not significant; they are consistent with the text-only result in that images do not substitute for coordinates.

D.3 Failure analysis

Termination labels name the stopping rule that fired, not the cause of failure. Rejected actions never end an episode, although they consume turns; the cycling rule stops an episode when the same fold is played from the same position three times. In the ablations, 191 of the 305 unsuccessful completed episodes (E_{id}; 300 under E_{plane}) end under the cycling rule (Table 12); in the earlier 447-attempt GPT-5.6 Luna snapshot, 215 of 384 do. These counts describe how episodes stopped, not why they failed, and we do not attribute failures to capabilities from them.

The per-turn logs of the image ablation episodes were not retained; the released runs keep each episode's accepted sequence and final state. Per-turn logs survive for the text-only GPT-5.6 Luna episodes, and we analysed those turn by turn (Table 15), replaying each accepted state and comparing it with the target. Each unsuccessful episode is assigned one of four mechanical categories (Appendix B.14): F1, mostly geometrically illegal proposals; F2, mostly legal folds off the target pattern; F3, the full pattern at the target depth but a wrong stack; F4, never reaching the target depth. With basic tools, 82% of proposed folds are rejected, mostly as off-pattern, and 29 of 36 failures are F2; with legal enumeration, the rejection rate falls to 13% and failures shift to F4 (21 of 32) and F3 (6 of 32). These failure counts use the outcomes as submitted (E_{plane}), as in Table 15. Once the tool removes illegal and off-pattern moves, the remaining failures are episodes that never reach the target depth, or reach it with the layers in the wrong order. The categories locate where an episode broke down, not which internal capability failed, and they come from one model without images; we have not verified that they transfer to the image ablations.

Reaching the target without stopping. An episode could fail by passing through the target state and folding on instead of finishing. Replaying every accepted state of the 143 saved episodes with per-turn logs and checking CP match plus E_{id} after each, an automatic stop at the first match would raise solves from 25 to 27. Failures are therefore almost never missed stops; in all but two unsuccessful episodes the model never reaches the target.

Exploration. We counted the distinct folded states (flat sheet excluded) each saved episode reaches and compared them with 20 random episodes of 80 legal folds on the same instance. The

median per-instance ratio of model to random distinct states (easy, some-layers) is 0.11 and 0.12 with text-only B, 0.16 and 0.24 with text-only L, and 0.25 and 0.18 with text-only L+A; it is lowest for images-only B (0.12 and 0.07). The model reaches between a tenth and a quarter of the states random folding reaches, and more states with enumeration than without, as with solve rate. With basic tools 72–81% of proposals are rejected; with legal enumeration 57–65% of accepted folds return to a state already reached. Part of the gap reflects episode length: model episodes stop after 20–45 turns on average (cycling rule or `finish`), whereas every random episode uses all 80 folds. Together with the matched baselines, this indicates that the model explores far less of the legal state space than undirected search and revisits what it has seen, rather than failing at the final step.

Backtracking. We examined every backtrack (a `go_to_step`, `restore_revision` or `remove_fold` that shortens the sequence) in the saved easy and some-layers episodes, and for each asked whether a local repair existed: whether the target is reachable (CP match plus E_{id}) from the sequence with its last one, two or three folds removed, by depth-limited search within the reference depth plus two (node cap 4,000). In the 120 text-only GPT-5.6 Luna L+A episodes (the original run and the five repeats) there are 384 backtracks; 104 (27%) return to the flat sheet, and the mean backtrack removes 2.6 folds. The search decided all 384: a repair within three folds existed for 232 (60%). A repair existed for 71 of the 104 returns to the flat sheet (68%) and for 161 of the 280 shorter backtracks (58%), so the model restarts about as often when a short repair is available as when it is not. In 21 Claude Sonnet 5.5 L+A episodes, 16 of 82 backtracks (20%) return to the flat sheet; a repair existed for 8 of these 16 (50%) and for 36 of the other 66 (55%).

After a backtrack of two or more folds, we asked whether the next accepted folds re-make the discarded ones, reaching the same states in the same order, and compared this with a policy that picks uniformly among the distinct legal folds at each state. With legal enumeration the models re-make the discarded folds less often than chance: GPT-5.6 Luna L+A in 32 of 293 cases against 61 expected, and Claude Sonnet 5.5 in 0 of 58 against 8.7 expected. With basic tools, where no list is shown, GPT-5.6 Luna re-makes them more often than chance (4 of 6 against 0.9 expected with B; 8 of 14 against 3.7 with B+A), but these counts are small.

The two families differ in whether they return to branches they have already given up. Counting every state on a discarded suffix, and every state reported to have no legal fold, as abandoned, 37% of GPT-5.6 Luna L+A add_folds after an episode's first backtrack land on an abandoned state again (340 of 931; 65 of them on a state with no legal fold), against 11% for Claude Sonnet 5.5 (26 of 243; 5 on such states); 77 of 120 Luna episodes and 7 of 21 Sonnet episodes contain such a revisit. Each turn is a fresh request, so the only record of abandoned branches is the action history in the prompt; Sonnet's lower rate is consistent with it using that history to avoid branches already tried, but its reasoning text is not returned by the interface we used, so we cannot tell how. Re-entering an abandoned shallow state is not always an error, since a solution may pass through it; re-entering a state with no legal fold always is.

The tier-3 comparison can mark a state as close to the target when no short repair exists. On `layers-0003`, GPT-5.6 Luna (nine restarts in six episodes) and Claude Sonnet 5.5 (one) restarted from a state with 8 of 9 layers, 6 of the 8 matching the target; in seven of these ten cases the search found no repair within three folds (in the other three a one-fold repair existed), and in the Sonnet case the state had no legal fold. On `easy-0002`, five Luna episodes and one Sonnet episode reached the target's 24 layers in the wrong order and restarted; none had a repair within three folds. Restarting from such states is therefore not by itself an error. What the episodes show is that the models do not distinguish a recoverable near miss from a trap: whether a short repair exists barely changes how often they restart, while the feedback's closeness signal (layer count and matching layers) also marks unrecoverable states as close. These counts come from a bounded search; they describe behaviour, not its cause.

A second model family. Claude Sonnet 5.5 ran the same text-only L+A configuration on the 20 instances of the repeated-trial set, one episode each (the first four easy episodes used an earlier prompt order). Under E_{id} it solves 5 of 10 easy and 9 of 10 some-layers instances (14/20); under E_{plane} it solves 7 of 10 easy, and the two E_{plane}-only solves ended with `finish` after the comparison tool reported *matches*, as for GPT-5.6 Luna (Section 6). The unsolved some-layers instance, `layers-0006`, is a protocol failure: in two attempts a single call exceeded the 300-second timeout (at turns 31 and 37); GPT-5.6 Luna never solved it in five repeats. On the same 20 instances

GPT-5.6 Luna solves 8.4 per repeat on average (2.4 easy, 6.0 some-layers) and 14 in at least one of five repeats. One Sonnet episode per instance thus matches Luna's five-repeat coverage, but the solved easy instances differ (Sonnet: 0001, 0004, 0006, 0009, 0010; Luna in at least one repeat: 0002, 0005, 0006, 0009, 0010). With one episode per instance and 20 instances this is consistent with a stronger model, not a measured difference; Sonnet still falls short of random legal folding on the same groups (Table 4).

The broader evaluation, ablations, BFS snapshot and CP-distance scan cover different attempt sets with different budgets; we report their denominators separately and do not pool them into a single controlled comparison. The one matched comparison of models and search is Table 4, on the 40 ablation instances.

E EXTENDED LIMITATIONS AND REPEATED-TRIAL ACCOUNTING

The main ablations cover Luna models and BFS, with an added 20-instance Sonnet evaluation, and the ablations use ten samples per group and configuration. This study is currently privately funded, and these limits follow from the cost of evaluation: with full image history, one Luna episode consumed about 3.1 million input tokens on average over 20 episodes. In pilots, Qwen3.5-9B (via Tinker) cycled without solving either of two samples, and one 80-turn GPT-6 Astra episode cost about US$400. We are seeking research funding to extend the evaluation to further model families and to the full corpus.

The action space covers all-layers and selected some-layers simple folds, excluding pre-creasing; the some-layers corpus currently extends to six reference folds. The environment represents zero-thickness flat states and does not model material mechanics or continuous collisions during folding. CP2Seq is a planning benchmark over an exact state space, not a model of physical folding, and we draw no engineering conclusions from it.

Depth and layer count are confounded in the all-layers tier. Whole-stack folds sampled uniformly almost always cross every layer, so the released medium and hard groups have medians of 94 and 404 final layers (maximum 16,384), and 96 of the 100 hard samples end with a footprint under 5% of the sheet. Results on these groups, including every model's and search's failure on hard, cannot be attributed to sequence depth alone. The generator already provides a cap on the number of layers a single fold may cross; with a cap of 16, all three groups fill their quotas with medians of 24, 60 and 76 final layers (maximum 120). The results in this paper are on the uncapped release; re-evaluation on the capped corpus is future work.

One episode per instance. The ablations run one episode per instance and configuration. Five further repeats of the text-only GPT-5.6 Luna L+A run solved 2, 3, 1, 3 and 3 easy instances under E_{id}, against 5 in the original run, with different instances solved. Both the totals and the set of solved instances therefore carry run-to-run variance that one episode cannot estimate; we interpret only differences in totals, with their uncertainty, and not per-instance differences between configurations. Repeated episodes per instance were then run for the text-only GPT-5.6 Luna L+A configuration: five episodes on each of the 10 easy and 10 some-layers instances (100 episodes). Under E_{id} the five repeats solve 7, 10, 6, 10 and 9 of 20 (42/100 episodes), and 14 of 20 instances are solved in at least one repeat (easy 5/10, some-layers 9/10); six instances are never solved. Under E_{plane} the counts are 47/100 episodes and 16/20 instances (easy 7/10). The five episodes counted as solved only under E_{plane} (`easy-0004` in four repeats, `easy-0003` in one) each ended with the model calling `finish` after the comparison it was shown reported *matches* with every layer agreeing, so they record the plane-only comparison tool, not a model error. The gap between per-episode rate and coverage measures run-to-run variance; it is obtained with five times the episode budget and is not evidence of more efficient search.

We treat these repeated trials as a stability analysis, not a causal diagnosis of failure. A planned diagnostic study freezes the task, state, history and reasoning effort, then varies one form of support: a summary of previously observed attempts, explicit action semantics, or uniform one-step previews. Its controls share the same legal-action menu. Such interventions can test the effect of support on decisions and subsequent success; they do not by themselves identify an internal memory or reasoning deficit. This study has not yet been run.

Input modality. Removing rendered images does not lower solve rate, and an images-only condition solves 1/40 with basic tools. The benchmark as configured measures reasoning over exact coordinates; it is not a test of visual perception.

Termination labels record stopping conditions. The trajectory categories of Section 5.3 locate where an episode broke down, but they are mechanical rules over observable states and do not identify which internal capability failed. Revisit and budget limits constrain the observed search, and the effect of a looser cycling limit is untested. Per-turn logs of the image ablations were not retained, so the trajectory analysis covers the text-only runs only, and the comparison-tool defect cannot be audited call by call there. Training overlap with the public corpus is unknown.

Corpus replay checks assess consistency with the shared generation engine. The submitted results were scored with a plane-only terminal comparator that did not check original-sheet region identity; the identity audit (Appendix A.7) found accepted solutions that fail the identity-aware relation, and all results are now re-scored under it. The comparison tool shown to models still uses the plane-only relation, so its *matches* can be wrong in the same way; the affected calls are counted in Appendix B.13.